

Electrodynamics

FY549

Lecture Notes for 3rd semester physics and physics and
engineering students

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*Based on D.J. Griffiths, Introduction to Electrodynamics
and D. Tong, Lectures on Electromagnetism*

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How to read these notes:

- Sections in **blue boxes** contain extra material that is helpful or interesting but not strictly necessary.
- Sections in **purple boxes** are for the interested reader who wants to go deeper.
- Sections in **red boxes** are in-class exercises.
- Key equations are marked with **orange boxes**.
- Text marked in **teal** refers to slides shown in the lecture.
- You have plenty of space in the left margin for your own comments.

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1 Maxwell's Equations

Literature: Griffiths Chapter 7.3

Maxwell's equations describe how electric and magnetic fields are generated from charges and currents. Historically, they are the culmination of theory to describe experimental results obtained by Faraday, Ampère, and many others. They form the axioms of classical electrodynamics.

Once we have written them down, we could stop right there — they encode (almost) everything in electrodynamics. We will study their consequences throughout this course, with emphasis on theory. Let us first review them and some of the vector calculus we need to understand them.

We write them in differential form and derive the integral form that you should be familiar with from your electrostatics course.

Consider the electric field $\mathbf{E} = (E_x, E_y, E_z)^T$ and the magnetic field $\mathbf{B} = (B_x, B_y, B_z)^T$.

1.1 Gauss's Law

$$\nabla \cdot \mathbf{E}(\mathbf{x}, t) = \frac{\rho(\mathbf{x}, t)}{\varepsilon_0} \quad (1)$$

Here, the left-hand side is the *divergence* of $\mathbf{E}$, with the nabla operator

$$\nabla = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix} \quad (2)$$

so that

$$\nabla \cdot \mathbf{E} = \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} = \sum_i \partial_i E_i \equiv \partial_i E_i \quad (3)$$

where in the last step we have used the *Einstein summation convention*: repeated indices are summed over.

Note on index notation

We will frequently use index notation where $\partial_i \equiv \frac{\partial}{\partial x_i}$ with $x_1 = x$, $x_2 = y$, $x_3 = z$. Repeated indices (one upper, one lower, or two of the same in this context) are implicitly summed over: $\partial_i E_i \equiv \sum_{i=1}^3 \partial_i E_i$.

In the second part of the course, it will also become important if we write the indices as subscripts or as superscripts, but this will not bother us now.

Note that $\mathbf{E}$ is a *vector field*: its strength and direction may vary with position and time. The divergence measures whether $\mathbf{E}$ has sources or sinks at a given point.

An example of a vector field with divergence is the electric field of a point charge, see the [slides for an example](#). Another example is the velocity field of a fluid flowing out of a faucet. We will explore this further in the exercises.

Gauss's law states: the source of divergence of $\mathbf{E}$ is the charge density:

- ρ : charge density (charge per unit volume)
- ϵ_0 : vacuum permittivity (ability of $\mathbf{E}$ to permeate vacuum)

$$\epsilon_0 = 8.85 \times 10^{-12} \text{C}^2 \text{N}^{-1} \text{m}^{-2} \quad \left(\epsilon_0 = \frac{10^7}{4\pi c^2} \right) \quad (4)$$

1.1.1 Physical Interpretation: Integral Form

Integrate both sides of Eq. (1) over a volume V :

$$\underbrace{\int_V (\nabla \cdot \mathbf{E}) dV}_{\text{Gauss' theorem}} = \frac{1}{\epsilon_0} \int_V \rho dV \quad (5)$$

$\searrow$
 $= \oint_S \mathbf{E} \cdot d\mathbf{a}$

The right-hand side gives the total charge Q enclosed in V .

Note on notation for volume element dV

Throughout the course, we will use dV interchangeably with d^3x to denote the volume element (Griffith uses $d\tau$). In Cartesian coordinates, this is $d^3x = dV = dx dy dz$.

Applying the *divergence theorem* (Gauss's theorem) to the left-hand side:

$$\oint_S \mathbf{E} \cdot d\mathbf{a} = \frac{Q_{\text{enc}}}{\epsilon_0} \quad (6)$$

where S is the closed surface enclosing V and $d\mathbf{a}$ points outward. The left-hand side is the *electric flux* through S .

The larger the charge inside V , the larger the flux of $\mathbf{E}$ through the closed surface S .

Note on flux

Flux is not flow. Flow is the average normal component of the field times the surface area.

Importantly, the *shape* of S does not matter — only the total charge enclosed in S does, see Figure 1.

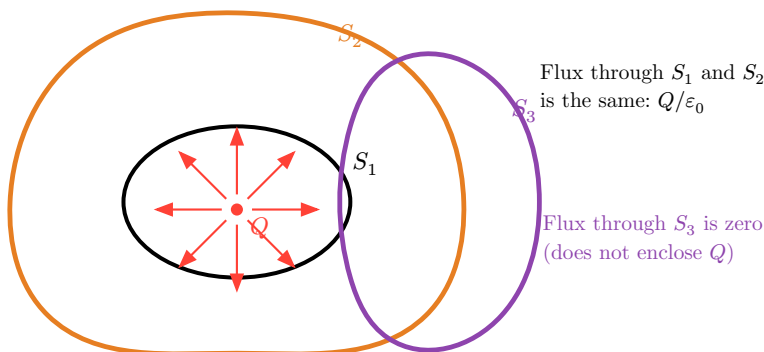


Figure 1: The flux through S_1 and S_2 is the same ($= Q/\epsilon_0$), since both enclose Q . The flux through S_3 is zero: as much flux enters V_3 as leaves it, since S_3 does not enclose the charge.

1.2 Gauss's Law for Magnetism

$$\nabla \cdot \mathbf{B} = 0 \quad (7)$$

This is analogous to Gauss's law in Eq. (1), but for the $\mathbf{B}$ field. It states that there are no magnetic charges (no magnetic monopoles — at least, we have not found them yet).

The net magnetic flux through any closed surface is zero: any outward flux is balanced by inward flux. If you like to visualize the fields with field lines: they are always *closed* for magnetic fields.

This is also true in the static case (magnetostatics).

1.3 Faraday's Law of Induction

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad (8)$$

This is Faraday's law of induction. Note that this is a *vector* equation (in contrast to Gauss's law), so both left- and right-hand sides have three components.

The curl is computed as

$$\nabla \times \mathbf{E} = \det \begin{pmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial_x & \partial_y & \partial_z \\ E_x & E_y & E_z \end{pmatrix} = \begin{pmatrix} \partial_y E_z - \partial_z E_y \\ -\partial_x E_z + \partial_z E_x \\ \partial_x E_y - \partial_y E_x \end{pmatrix} \quad (9)$$

The Levi-Civita tensor

In index notation, using the Levi-Civita tensor:

$$(\nabla \times \mathbf{E})_i = \varepsilon_{ijk} \partial_j E_k \quad (10)$$

The Levi-Civita tensor ε_{ijk} is defined as

$$\varepsilon_{ijk} = \begin{cases} +1 & \text{even permutation of } 1, 2, 3 \\ -1 & \text{odd permutation of } 1, 2, 3 \\ 0 & \text{for repeated indices} \end{cases} \quad (11)$$

An *even* permutation is one obtained by cyclic rotation: $123 \rightarrow 231 \rightarrow 312$. You can think of a cyclic rotation as a rotation of a triangle with the numbers 1, 2, 3 at the corners. An *odd* permutation is obtained by swapping two indices: $123 \rightarrow 213 \rightarrow 321 \rightarrow 132 \rightarrow 231 \rightarrow 312$.

1.3.1 Physical Interpretation: Integral Form

Integrate both sides over a surface S :

$$\underbrace{\int_S (\nabla \times \mathbf{E}) \cdot d\mathbf{a}}_{\text{Stokes' theorem}} = - \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{a} \stackrel{\text{Holding } S \text{ constant in time}}{=} - \frac{d}{dt} \underbrace{\int_S \mathbf{B} \cdot d\mathbf{a}}_{\text{Magnetic flux } \Phi_B \text{ through surface } S} \quad (12)$$

$$\stackrel{\text{Stokes' theorem}}{=} \oint_C \mathbf{E} \cdot d\mathbf{l}$$

where C is the closed curve bounding S , and $\Phi_B = \int_S \mathbf{B} \cdot d\mathbf{a}$ is the magnetic flux through the surface. The last equality only holds when S is fixed in time, i.e. the surface does not move or deform as time progresses. In that case, the time derivative can be taken outside of the integral.

Collecting results gives the integral form of Faraday's law:

$$\oint_C \mathbf{E} \cdot d\mathbf{l} = -\frac{d\Phi_B}{dt} \quad (13)$$

Geometrical interpretation of Stokes' theorem

The curl measures the “twist” of a vector field (a whirlpool is a region with high curl). The integral of the flux of the curl through a surface represents the total “amount of swirl”. Stokes' theorem says this equals the circulation — the flow along the boundary. This is illustrated in Figure 2. Crucially, the integral depends only on the boundary C , not on the surface S itself.

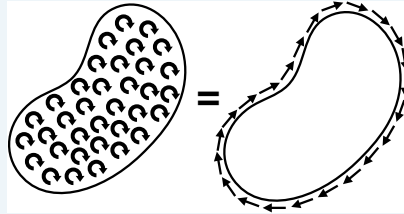


Figure 2: Small circulation loops covering the surface cancel along shared interior edges, leaving only the circulation along the outer boundary C .

The physical content: a time-varying magnetic flux through a surface induces an electric field along the closed path around the surface.

If you place a wire along this path, current will flow as $\mathbf{E}$ exerts force on the charge carriers inside the wire. This is the *electromotive force* (emf):

$$\mathcal{E} = \oint \mathbf{f} \cdot d\mathbf{l} = -\frac{d\Phi_B}{dt} \quad (14)$$

where $\mathbf{f}$ is the force per unit charge responsible for pushing the charge around the circuit.

Lenz's law

The minus sign in Faraday's law encodes Lenz's law: the emf opposes the change in magnetic flux.

This was Faraday's experimental breakthrough: three experiments involving a wire and a magnetic field all yielded a current through the wire, [see the slides for the experimental setups](#). In experiment (a), the wire moves — this is a case of *motional emf*, where the relevant force is the Lorentz force $\mathbf{f} = \mathbf{v} \times \mathbf{B}$. Experiment (b) gives the same result — only relative motion matters. In experiment (c), the wire is not moving ($\mathbf{v} = 0$), so who does the work? The answer: the electric field that is *induced* by the changing magnetic field. This was Faraday's great insight.

1.4 Ampère's Law with Maxwell's Addition

$$\underbrace{\nabla \times \mathbf{B} = \mu_0 \mathbf{J}}_{\text{Ampère's law}} + \underbrace{\mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}}_{\text{Maxwell's addition}} \quad (15)$$

The first term is *Ampère's law*, and the second is *Maxwell's addition* (the displacement current term).

Here:

- $\mathbf{J}$: current density — the amount of current passing through a surface per unit time. The number of charges passing per second through an area element $d\mathbf{S}$ is $\mathbf{J} \cdot \hat{\mathbf{n}} dS$. Units: $[\mathbf{J}] = \text{A} \cdot \text{m}^{-2}$.
- μ_0 : permeability of free space, $\mu_0 \approx 4\pi \times 10^{-7} \text{ N A}^{-2}$.

Definition of μ_0

μ_0 is defined through the force of attraction between two infinite wires with infinitesimal cross section, 1 m apart, carrying 1 A current in the same direction. In natural units: $\mu_0/(4\pi) = \alpha/(2\pi) \cdot \hbar/(e^2 c)$. Here, α is the fine-structure constant, $\hbar$ is the reduced Planck constant (see your quantum mechanics class!), e is the electric charge and c is the speed of light.

1.4.1 Physical Interpretation: Integral Form (Static Case)

In the static case ($\frac{\partial \mathbf{E}}{\partial t} = 0$), integrate over a surface S and apply Stokes' theorem:

$$\int_S (\nabla \times \mathbf{B}) \cdot d\mathbf{a} = \oint_C \mathbf{B} \cdot d\mathbf{l} = \mu_0 \int_S \mathbf{J} \cdot d\mathbf{a} \quad (16)$$

The right-hand side is μ_0 times the total current through S . The direction of C is determined by the right-hand rule with respect to the surface normal $\hat{\mathbf{n}}$.

The magnetic field along C is determined by the total current passing through S .

1.4.2 Why Is Maxwell's Addition Necessary?

The displacement current term ensures the *consistency* of Maxwell's equations.

Take the divergence of Faraday's law Eq. (8):

$$\underbrace{\nabla \cdot (\nabla \times \mathbf{E})}_{=0 \text{ (divergence of curl always zero)}} = -\frac{\partial}{\partial t} \left(\underbrace{\nabla \cdot \mathbf{B}}_{=0 \text{ (2nd Maxwell equation)}} \right) = 0 \quad \checkmark \quad (17)$$

Now take the divergence of Ampère's law in Eq. (15) *without* Maxwell's addition:

$$\underbrace{\nabla \cdot (\nabla \times \mathbf{B})}_{=0} = \mu_0 \nabla \cdot \mathbf{J} \quad (18)$$

This would require $\nabla \cdot \mathbf{J} = 0$ always, which is not true in general (e.g. when charges leave a volume).

With Maxwell's addition:

$$0 = \mu_0 \nabla \cdot \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial}{\partial t} (\nabla \cdot \mathbf{E}) = \mu_0 \nabla \cdot \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial}{\partial t} \frac{\rho}{\varepsilon_0} \quad (19)$$

which gives the *continuity equation*:

$$\nabla \cdot \mathbf{J} + \frac{\partial \rho}{\partial t} = 0 \quad (20)$$

This is an important result that we will use often. It expresses *conservation of charge*: a change in charge density in time is associated with a divergence in the current (charges flowing in or out of the volume).

In integral form:

$$\oint_S \mathbf{J} \cdot d\mathbf{a} = -\frac{dQ}{dt} \quad (21)$$

1.5 Summary: Maxwell's Equations

These are Maxwell's *microscopic* equations: ρ and $\mathbf{J}$ include *all* charges and currents, also “internal” ones like bound charges in atoms and molecules. They are a system of coupled first-order partial differential equations.

In-class exercise

- How many equations do we have (counting components)?
- Why didn't we write them in Cartesian coordinates?

Here they are in one place (combining Eq. (1), Eq. (7), Eq. (8), and Eq. (15)):

$$\begin{aligned} \nabla \cdot \mathbf{E} &= \frac{\rho}{\varepsilon_0} & (i) \\ \nabla \cdot \mathbf{B} &= 0 & (ii) \\ \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} & (iii) \\ \nabla \times \mathbf{B} &= \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} & (iv) \end{aligned} \quad (22)$$

1.5.1 Consequences

Suppose you switch on a current $\mathbf{J}$:

- $\mathbf{J}$ varies in time
- generates a time-variable $\mathbf{B}$ field through Ampère's law (iv)
- generates a time-variable $\mathbf{E}$ field through Faraday's law (iii)
- which in turn modifies $\mathbf{B}$, and so on...

This sounds like an oscillation that propagates — Maxwell's equations are the source of *electromagnetic waves*. We will get to that in the coming weeks.

2 Maxwell's Equations in Matter

Literature: Griffiths Ch. 7.3 (cont'd)

Maxwell's equations Eq. (22) (i)–(iv) are also valid in the presence of matter. However, ρ and $\mathbf{J}$ then include *all* charges and currents, also those that are “bound” in matter — over which we have no direct influence.

It is convenient to rewrite Maxwell's equations in terms of new fields $\mathbf{D}$ and $\mathbf{H}$ that distinguish between *free* charges and currents (ρ_f , $\mathbf{J}_f$) that we can experimentally control, and *bound* charges and currents (ρ_b , $\mathbf{J}_b$).

The bound charges ρ_b and currents $\mathbf{J}_b$ are produced as $\mathbf{E}$ and $\mathbf{B}$ *polarize* and *magnetize* a medium (e.g. electrons and protons in a material get slightly shifted against each other).

At the microscopic level, $\mathbf{E}$ and $\mathbf{B}$ are extremely complex: close to particles, fields become extremely large; further away, closer to another particle, the field might flip direction; at finite temperature, particles undergo thermal fluctuations and fields change continuously.

We focus on the *macroscopic* level — averaged over large enough regions such that we are not affected by microscopic effects. This approach of finding relevant information using an *effective* (averaged) description instead of accounting for all microscopic physics is ubiquitous in physics.

2.1 Dielectrics and Polarization

In-class exercise

What happens inside matter when you apply an electric field?

Answer: it depends on the material.

In a **conductor**, there are lots of free electrons. Any electric field is compensated very fast, so $\mathbf{E} = 0$ inside the conductor. (There will still be a current when you hook up a battery, but inside the wire $\mathbf{E} = 0$. We will look at conductors later.)

In **dielectrics** (non-conducting materials), there are no free charge carriers — electrons are bound in atoms. When you apply an electric field, the positions of electrons (negatively charged) and nuclei (positively charged) shift (in which direction will the particles shift with respect to each other?), leading to small *dipole moments*:

$$\mathbf{p} = q \cdot \mathbf{d} = \alpha \cdot \mathbf{E} \quad (23)$$

where α is the *polarizability* (a scalar or tensor).

Molecules like water that already have a dipole moment will experience a torque and align with the $\mathbf{E}$ field.

The result: lots of dipoles pointing along $\mathbf{E}$ — the material gets *polarized*. A convenient measure is the **polarization**:

$$\mathbf{P} = \text{dipole moment per unit volume}$$

2.1.1 Bound Charges

We can also understand polarization in terms of *bound charges* or the bound charge density ρ_b .

When the induced dipoles all align, the interior charges cancel in pairs, as depicted in Figure Figure 3. The net effect is a surface charge — what we call *bound charge*.

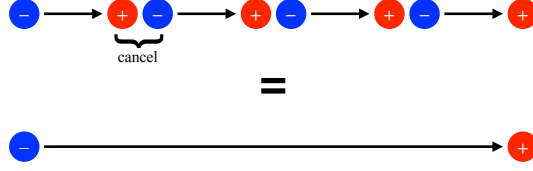


Figure 3: Illustration of aligned dipoles inside a material. The inner charges cancel leaving effectively a bound surface charge.

One can show (Griffiths Ch. 4) that bound charges and polarization are connected through

$$-\nabla \cdot \mathbf{P} = \rho_b \quad (24)$$

The total charge density is

$$\rho = \rho_b + \rho_f \quad (25)$$

where ρ_f is the *free* charge density that we can manipulate in the lab.

Example on Polarization

Imagine a sphere with a uniform distribution of positive and negative charges, as shown in Figure 4.

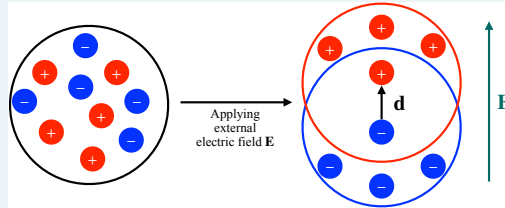


Figure 4: When an electric field is applied, the positive charges move in the direction of $\mathbf{E}$ while the negative charges move in the opposite direction. As a result, the spheres are shifted against each other.

The electric field in the overlapping region is given by

$$\mathbf{E}_{\text{overlap}} = -\frac{1}{4\pi\epsilon_0} \frac{q \cdot \mathbf{d}}{R^3}, \quad (26)$$

where R is the radius of the sphere (believe me or check in Griffiths Ch. 2 and 4).

The electric field can be expressed in terms of the polarization $\mathbf{P}$ replacing the dipole moment $\mathbf{p}$ (with charge q the net positive charge) with the polarization $\mathbf{P}$:

$$\mathbf{p} = q \cdot \mathbf{d} = \mathbf{P} \cdot V = \mathbf{P} \cdot \left(\frac{4}{3}\pi R^3 \right), \quad (27)$$

which is uniform accross the sphere. Plugging in:

$$\Rightarrow \mathbf{E}_{\text{overlap}} = -\frac{1}{3\varepsilon_0} \mathbf{P}. \quad (28)$$

Outside the sphere, for distances $|\mathbf{r}| \gg |\mathbf{d}|$ we have a dipole with potential

$$U = \frac{1}{4\pi\varepsilon_0} \frac{\mathbf{p} \cdot \mathbf{r}}{|\mathbf{r}|^3} \quad (29)$$

The total electric field is then the sum of the field that caused the polarization in the first place and the electric field produced by the polarizatio, i.e. the shifting of the spheres. The result is shown in Figure 5.

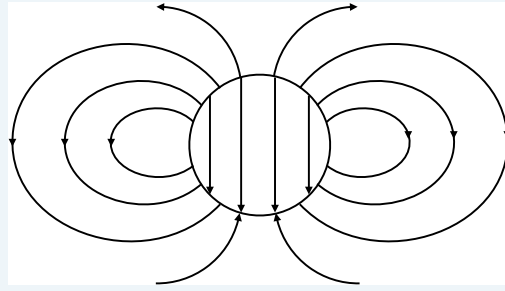


Figure 5: The total electric field from the uniform polarization. The field is uniform inside the sphere and behaves like a dipole outside.

2.1.2 The Electric Displacement $\mathbf{D}$

We define the *electric displacement* $\mathbf{D}$ whose divergence equals the free charge density:

$$\nabla \cdot \mathbf{D} = \rho_f \quad (30)$$

This is like Gauss's law but only for free charges and without ε_0 . Starting from the divergence of the total $\mathbf{E}$ field:

$$\begin{aligned} \nabla \cdot \mathbf{E} &= \frac{1}{\varepsilon_0} (\rho_f + \rho_b) \\ &= \frac{1}{\varepsilon_0} \nabla \cdot \mathbf{D} - \frac{1}{\varepsilon_0} \nabla \cdot \mathbf{P} \\ &= \frac{1}{\varepsilon_0} \nabla \cdot (\mathbf{D} - \mathbf{P}) \end{aligned} \quad (31)$$

This gives us the definition of $\mathbf{D}$:

$$\mathbf{D} = \varepsilon_0 \mathbf{E} + \mathbf{P} \quad (32)$$

This is actually the definition of $\mathbf{D}$, motivated here from consistency with Maxwell's equations.

Next lecture: we will see what happens in matter when you apply a magnetic field, and rewrite Maxwell's equations fully in terms of free charges and currents.

2.2 Magnetization

Last lecture we looked at the electric field in dielectrics and introduced the *electric displacement*

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}, \quad \nabla \cdot \mathbf{D} = \rho_f. \quad (33)$$

We saw that dielectrics become polarized when an external field is applied (note that $\mathbf{E}$ above is the *total* field).

We now ask the analogous question for magnetic fields.

In-class exercise

What happens when you place a piece of material in a magnetic field?

Again, it depends on the type of material.

Some materials, like iron, become magnets themselves and can even stay magnetized once the external field is switched off. Such substances (e.g., iron, nickel, cobalt) are called **ferromagnets** — we will not discuss them further in this course.

But other materials are also affected by external fields, and are divided into:

- **Paramagnets:** acquire a magnetization *parallel* to $\mathbf{B}$.
- **Diamagnets:** acquire a magnetization *antiparallel* to $\mathbf{B}$.

Compared to ferromagnets, the magnetization is much smaller (you can't pick up wood with a magnet).

Let's look at paramagnets and diamagnets more closely.

2.2.1 Microscopic Origin: Magnetic Dipoles

In fact, all magnetic phenomena are caused by tiny currents:

- electrons that have *spin* (imagine them as spinning charged spheres, even though this picture is not correct in a quantum-mechanical sense),
- electrons that *orbit* around the nucleus.

Both phenomena generate current loops that we can treat as magnetic dipoles, sketched in Figure 6. The dipole moment is given by

$$\mathbf{m} = I \mathbf{A} \quad (34)$$

where I is the current circulating around the loop of area A , and $\mathbf{A} = A\hat{\mathbf{n}}$ points along the loop's normal (direction set by the right-hand rule).

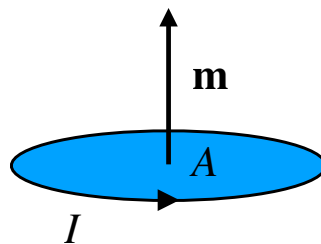


Figure 6: A magnetic dipole moment $\mathbf{m}$ is generated by a current loop. The direction of $\mathbf{m}$ is given by the right-hand rule.

2.2.2 Paramagnetism

Without an external field, these dipoles point in random directions. Once you apply a field, the dipoles experience a torque that will line them up parallel to the field. This torque accounts for paramagnetism.

You might expect this to be a universal phenomenon, since every spinning electron constitutes a dipole.

However, in quantum mechanics we have the **Pauli exclusion principle**, which states that two electrons (or more generally, *fermions* — particles with half-integer spin) cannot occupy the same quantum state. This tends to lock electrons with opposite spin together in atoms, so that their torques neutralize.

As a result, paramagnetism occurs mostly in atoms or molecules with an *odd* number of electrons. But even in such materials, random thermal fluctuations compete with the ordering of the dipoles.

2.2.3 Diamagnetism

Orbits of electrons also constitute tiny current loops, but these are randomly oriented. In the presence of a field, an orbiting electron experiences an additional Lorentz force that will either speed it up or slow it down. This changes its dipole moment *antiparallel* to the field. This is diamagnetism.

Diamagnetism is typically weaker than paramagnetism, and mainly observed in atoms/molecules with an *even* number of electrons (where paramagnetism is largely cancelled by the Pauli exclusion principle).

2.2.4 The Magnetization M

Similarly to polarization, we can thus define the **magnetization**:

$$\mathbf{M} = \text{magnetic dipoles per unit volume}$$

The magnetization is connected to a **bound current density** $\mathbf{J}_b$ (produced by the electron spins and orbits) through

$$\mathbf{J}_b = \nabla \times \mathbf{M} \quad (35)$$

[See the slides for an illustration.](#)

2.2.5 The Auxiliary Field $\mathbf{H}$

The currents we control directly — e.g. by increasing a voltage — are called **free currents**, and they produce a field $\mathbf{H}$, analogous to the electric displacement $\mathbf{D}$ in the electrostatic case:

$$\nabla \times \mathbf{H} = \mathbf{J}_f. \quad (36)$$

In practice, $\mathbf{H}$ is used far more frequently than $\mathbf{D}$. Usually $\mathbf{H}$ is defined through

$$\mathbf{H} = \frac{1}{\mu_0} \mathbf{B} - \mathbf{M} \quad (37)$$

Unlike $\mathbf{B}$, the field $\mathbf{H}$ is not generally divergence-free: $\nabla \cdot \mathbf{H} = -\nabla \cdot \mathbf{M}$, which need not vanish.

2.2.6 The Polarization Current

So far, the relation $\mathbf{J}_b = \nabla \times \mathbf{M}$ was only established for the electro-/magnetostatic case (only in this case we have $\mathbf{J} = \mathbf{J}_f + \mathbf{J}_b$). In the time-dependent case, we get an additional contribution to the current density: a change in polarization also drives a **polarization current** $\mathbf{J}_p$:

$$\mathbf{J}_p = \frac{\partial \mathbf{P}}{\partial t} \quad (38)$$

What's the physical meaning of this current? It has nothing to do with the bound current $\mathbf{J}_b$; $\mathbf{J}_b$ is associated with the magnetization of the material which involves the spin and orbital motion of electrons. In contrast, the polarization current $\mathbf{J}_p$ is the result of the linear motion of charge as the polarization changes. For example: if $\mathbf{P}$ points in the $+x$ direction and increases, positive charges will move slightly in the $+x$ direction while negative charges move slightly in the $-x$ direction. The cumulative effect of these tiny shifts is the current $\mathbf{J}_p$.

This current is essential to ensure conservation of bound charge. We can check this against the continuity equation (Eq. (20)):

$$\nabla \cdot \mathbf{J}_p = \nabla \cdot \frac{\partial \mathbf{P}}{\partial t} = \frac{\partial}{\partial t} (\nabla \cdot \mathbf{P}) = -\frac{\partial \rho_b}{\partial t}, \quad (39)$$

where we used the definition of the polarization current (Eq. (38)). In the last step we used the relation between polarization and bound charge, Eq. (24). This is exactly the continuity equation for the bound charge density ρ_b , as required.

Does a changing magnetization lead to a charge accumulation?

No — a changing magnetization changes the bound current, but

$$\nabla \cdot \mathbf{J}_b = \underbrace{\nabla \cdot (\nabla \times \mathbf{M})}_{=0 \text{ (divergence of curl always zero)}}, \quad (40)$$

so a changing $\mathbf{M}$ redistributes current but never accumulates net charge anywhere. (Note that this is not to be confused with the polarization current $\mathbf{J}_p$ above, which is associated with the polarization $\mathbf{P}$, not the magnetization.)

2.2.7 Ampère's Law in Matter

The total current density now has three contributions:

$$\begin{aligned} \mathbf{J} &= \mathbf{J}_f + \mathbf{J}_b + \mathbf{J}_p \\ &= \mathbf{J}_f + \nabla \times \mathbf{M} + \frac{\partial \mathbf{P}}{\partial t} \end{aligned} \quad (41)$$

Let's write Ampère's law with Maxwell's addition, Eq. (22) (iii), in terms of $\mathbf{H}$ and $\mathbf{D}$ instead of $\mathbf{B}$ and $\mathbf{J}$. Starting from

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}, \quad (42)$$

we substitute $\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M})$ (from Eq. (37)) on the left-hand side, and Eq. (41) on the right-hand side:

$$\underbrace{\mu_0 \nabla \times (\mathbf{H} + \mathbf{M})}_{\text{using definition of } \mathbf{H}} = \underbrace{\mu_0 \left(\mathbf{J}_f + \nabla \times \mathbf{M} + \frac{\partial \mathbf{P}}{\partial t} \right)}_{\text{using definition of } \mathbf{J}} + \underbrace{\mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}}_{\text{will combine into } \mathbf{D}} \quad (43)$$

The $\mu_0 \nabla \times \mathbf{M}$ term appears on both sides and cancels. Dividing by μ_0 :

$$\nabla \times \mathbf{H} = \mathbf{J}_f + \frac{\partial \mathbf{P}}{\partial t} + \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}. \quad (44)$$

Using the definition of $\mathbf{D}$, Eq. (32), the last two terms combine into $\frac{\partial \mathbf{D}}{\partial t}$:

$$\nabla \times \mathbf{H} = \mathbf{J}_f + \frac{\partial \mathbf{D}}{\partial t} \quad (45)$$

Faraday's law Eq. (22) (iii) and Gauss's law for magnetism Eq. (22) (ii) stay unchanged — they don't involve currents or a free/bound charge distinction.

2.3 Macroscopic Maxwell's Equations

We arrive at an alternative formulation of Maxwell's equations, valid inside matter, sometimes called the **macroscopic** Maxwell's equations:

$$\begin{aligned} \nabla \cdot \mathbf{D} &= \rho_f & (i) \\ \nabla \cdot \mathbf{B} &= 0 & (ii) \\ \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} & (iii) \\ \nabla \times \mathbf{H} &= \mathbf{J}_f + \frac{\partial \mathbf{D}}{\partial t} & (iv) \end{aligned} \quad (46)$$

plus the constitutive relations

$$\mathbf{D} = \varepsilon_0 \mathbf{E} + \mathbf{P} \quad \text{and} \quad \mathbf{H} = \frac{1}{\mu_0} \mathbf{B} - \mathbf{M}. \quad (47)$$

Alternatively, one can stick with the *microscopic* Maxwell's equations (Eq. (22)) in terms of $\mathbf{E}$ and $\mathbf{B}$ only, as long as the sources ρ and $\mathbf{J}$ include *all* charges and currents — free and bound.

2.4 Linear Media and Susceptibility

Often, the displacement and magnetizing fields are written in terms of the **permittivity** ε and **permeability** μ of the material:

$$\mathbf{D} = \varepsilon \mathbf{E}, \quad \mathbf{H} = \frac{1}{\mu} \mathbf{B}. \quad (48)$$

ε and μ can be expressed in terms of the electric and magnetic **susceptibilities** χ_e , χ_m :

$$\varepsilon = \varepsilon_0(1 + \chi_e), \quad \mu = \mu_0(1 + \chi_m). \quad (49)$$

The dielectric constant

The (relative) **dielectric constant** is

$$\varepsilon_r = \frac{\varepsilon}{\varepsilon_0} = 1 + \chi_e. \quad (50)$$

χ_e and χ_m are dimensionless numbers, [see the slides for typical values](#).

Linear media have a linear relation between polarization/magnetization and the fields:

$$\mathbf{P} = \varepsilon_0 \chi_e \mathbf{E}, \quad \mathbf{M} = \chi_m \mathbf{H}. \quad (51)$$

- For **homogeneous** materials, ε and μ are constants; for **inhomogeneous** materials, they are functions of position.
- For **isotropic** materials, ε and μ are scalars; for **anisotropic** materials (like crystals), they are tensors.

Interlude: Connection to my research on axions

Axions could solve two pressing problems in particle physics and cosmology:

- The nature of *dark matter*
- The non-observation of an electric dipole moment of the neutron (the *strong CP problem*)

If axions existed, they could couple to photons in the presence of $\mathbf{B}$ -fields. If axions are dark matter, they would surround us and *modify Maxwell's equations*. Gauss's law would get an additional source term from the spatial gradient of the axion field:

$$\rho \rightarrow \rho - \frac{g_{a\gamma}}{\mu_0 c} \mathbf{B} \cdot \nabla a \quad (52)$$

where $g_{a\gamma}$ is the axion-photon coupling constant and a the axion field. Ampère's law similarly gets modified:

$$\mathbf{J}_f + \frac{\partial \mathbf{D}}{\partial t} \rightarrow \mathbf{J}_f + \frac{\partial \mathbf{D}}{\partial t} + \frac{g_{a\gamma}}{\mu_0 c} \left[\frac{\partial a}{\partial t} \mathbf{B} - \mathbf{E} \times \nabla a \right] \quad (53)$$

Even without charges and currents, you could generate electromagnetic fields through axion dark matter! Typically, the spatial variations of the axion field are negligible, so the gradient term can be dropped leaving the axion current as the new dominating source term,

$$\mathbf{J}_a = \frac{g_{a\gamma}}{\mu_0 c} \frac{\partial a}{\partial t} \mathbf{B}. \quad (54)$$

The axion field oscillates at a frequency $\omega_a = m_a \frac{c^2}{\hbar}$ set by the axion mass m_a , which is unknown but expected to be very small. This leads to an oscillating

axion current in the presence of a static magnetic field, which is the basis for many axion dark matter searches also here at SDU.

2.5 Boundary Conditions

Starting from Maxwell's equations in matter, we now ask: what happens at the boundary between two media with different permittivities $\varepsilon_1, \varepsilon_2$ and permeabilities μ_1, μ_2 ? For now we only look at the general boundary conditions that the electric and magnetic fields must fulfil.

We will soon see that electromagnetic fields are propagating waves, and that these boundary conditions lead to reflection, transmission, and refraction (e.g. light passing from air into glass or water).

In general, $\mathbf{E}$, $\mathbf{D}$, $\mathbf{B}$, and $\mathbf{H}$ are all discontinuous at the boundary between two media.

2.5.1 Perpendicular Components: $\mathbf{D}$ and $\mathbf{B}$

The form of the discontinuity can be derived from Maxwell's equations in integral form. Starting from Gauss's law, (i) in Eq. (46), and applying the divergence theorem:

$$\int_V (\nabla \cdot \mathbf{D}) dV = \oint_S \mathbf{D} \cdot d\mathbf{a} = Q_{f, \text{enc}}, \quad (55)$$

where $Q_{f, \text{enc}}$ is the total free charge enclosed by the surface S . This holds for *any* closed surface — including a thin “pillbox” straddling the boundary between the two media, see Figure 7.

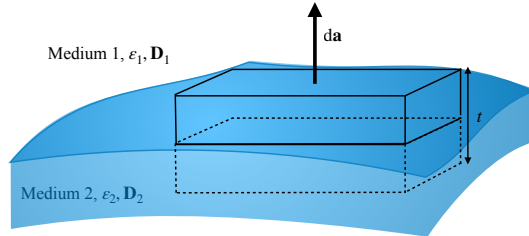


Figure 7: A thin pillbox straddling the boundary between two media. The outward normal vectors $\hat{\mathbf{n}}$ point into medium 1 and medium 2, respectively.

Consider the limit where the pillbox thickness $t \rightarrow 0$: the integral only gets a contribution from the top and bottom faces (each of area a , with outward normal vectors $\pm \hat{\mathbf{n}}$ pointing into medium 1 and medium 2 respectively):

$$\oint_S \mathbf{D} \cdot d\mathbf{a} = (\mathbf{D}_1 \cdot \hat{\mathbf{n}})a - (\mathbf{D}_2 \cdot \hat{\mathbf{n}})a = Q_{f, \text{enc}} = \sigma_f \cdot a, \quad (56)$$

where σ_f is the free surface charge density. Dividing by a :

$$D_1^\perp - D_2^\perp = \sigma_f \quad (57)$$

where $\perp$ denotes the component perpendicular to the boundary. For linear media (Eq. (48)) with no free surface charge:

$$\varepsilon_1 E_1^\perp = \varepsilon_2 E_2^\perp. \quad (58)$$

Similarly, from Gauss's law for magnetism, (ii) in Eq. (46), we have $\oint_S \mathbf{B} \cdot d\mathbf{a} = 0$. The same pillbox argument then gives:

$$B_1^\perp = B_2^\perp \quad (59)$$

The perpendicular component of $\mathbf{B}$ is always continuous.

2.5.2 Parallel Components: $\mathbf{E}$ and $\mathbf{H}$

Now we turn to the two Maxwell equations that involve curls. Using Stokes' theorem, for a closed path C around S , we find the integral forms of (iii) and (iv) in Eq. (46),

$$\int_S (\nabla \times \mathbf{E}) \cdot d\mathbf{a} = \oint_C \mathbf{E} \cdot d\mathbf{l}, \quad (60)$$

and similarly

$$\int_S (\nabla \times \mathbf{H}) \cdot d\mathbf{a} = \oint_C \mathbf{H} \cdot d\mathbf{l} = I_{f, \text{enc}} + \frac{d}{dt} \int_S \mathbf{D} \cdot d\mathbf{a}, \quad (61)$$

where $I_{f, \text{enc}}$ is the total free current enclosed by the loop.

Consider a thin rectangular Amperian loop with sides of length Δl straddling the boundary, with height h , see Figure Figure 8. In the limit $h \rightarrow 0$, the area S enclosed by the loop vanishes, so the flux term on the right-hand side of Faraday's law vanishes:

$$\mathbf{E}_1 \cdot \Delta \mathbf{l} - \mathbf{E}_2 \cdot \Delta \mathbf{l} = -\frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{a} \xrightarrow{h \rightarrow 0} 0, \quad (62)$$

so

$$\mathbf{E}_1^\parallel = \mathbf{E}_2^\parallel \quad (63)$$

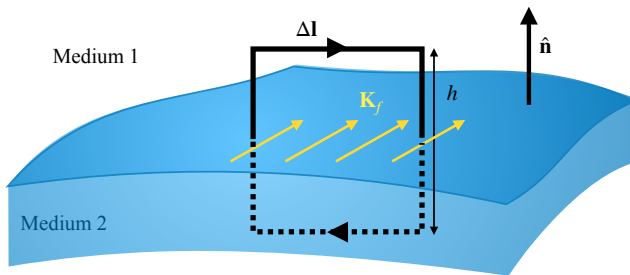


Figure 8: A thin rectangular Amperian loop straddling the boundary between two media.

The parallel component of $\mathbf{E}$ is always continuous across the boundary (note that $\mathbf{E}^\parallel$ is a two-dimensional vector, living in the plane of the boundary).

For Ampère's law (Eq. (61)), the total free current enclosed by the loop can be written in terms of a free surface current density $\mathbf{K}_f$:

$$I_{f, \text{enc}} = \mathbf{K}_f \cdot (\hat{\mathbf{n}} \times \Delta \mathbf{l}), \quad (64)$$

where $\hat{\mathbf{n}} \times \Delta \mathbf{l}$ is the vector perpendicular to the loop, so that $\mathbf{K}_f \cdot (\hat{\mathbf{n}} \times \Delta \mathbf{l})$ picks out the component of $\mathbf{K}_f$ passing through the loop. Using the cyclic property of the scalar triple product, $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B})$:

$$I_{f, \text{enc}} = \Delta \mathbf{l} \cdot (\mathbf{K}_f \times \hat{\mathbf{n}}), \quad (65)$$

so that

$$\mathbf{H}_1 \cdot \Delta \mathbf{l} - \mathbf{H}_2 \cdot \Delta \mathbf{l} = I_{f, \text{enc}} = \Delta \mathbf{l} \cdot \mathbf{K}_f \times \hat{\mathbf{n}}. \quad (66)$$

Pulling out the common factor $\Delta \mathbf{l}$, we find

 free surface current density

$$\mathbf{H}_1^{\parallel} - \mathbf{H}_2^{\parallel} = \mathbf{K}_f \times \hat{\mathbf{n}}. \quad (67)$$

The parallel component of $\mathbf{H}$ has a discontinuity proportional to the free surface current density. For linear media (Eq. (48)) with no surface currents:

$$\frac{1}{\mu_1} \mathbf{B}_1^{\parallel} = \frac{1}{\mu_2} \mathbf{B}_2^{\parallel}. \quad (68)$$

Summary: boundary conditions

Collecting the four results derived above (Eq. (57), Eq. (59), Eq. (63), and Eq. (67) respectively), the fields on either side of a boundary between two media are related by:

$$\begin{aligned} D_1^{\perp} - D_2^{\perp} &= \sigma_f & (\text{i, perpendicular}) \\ B_1^{\perp} &= B_2^{\perp} & (\text{ii, perpendicular}) \\ \mathbf{E}_1^{\parallel} &= \mathbf{E}_2^{\parallel} & (\text{iii, parallel}) \\ \mathbf{H}_1^{\parallel} - \mathbf{H}_2^{\parallel} &= \mathbf{K}_f \times \hat{\mathbf{n}} & (\text{iv, parallel}) \end{aligned} \quad (69)$$

For **linear media** (Eq. (48)) with no free surface charge or current ($\sigma_f = 0$, $\mathbf{K}_f = \mathbf{0}$), the perpendicular and parallel conditions on $\mathbf{D}$ and $\mathbf{H}$ simplify to relations purely between $\mathbf{E}$ - and $\mathbf{B}$ -fields (Eq. (58), Eq. (59), Eq. (63), and Eq. (68) respectively):

$$\begin{aligned} \varepsilon_1 E_1^{\perp} &= \varepsilon_2 E_2^{\perp} & (\text{i, perpendicular}) \\ B_1^{\perp} &= B_2^{\perp} & (\text{ii, perpendicular}) \\ \mathbf{E}_1^{\parallel} &= \mathbf{E}_2^{\parallel} & (\text{iii, parallel}) \\ \frac{1}{\mu_1} \mathbf{B}_1^{\parallel} &= \frac{1}{\mu_2} \mathbf{B}_2^{\parallel} & (\text{iv, parallel}) \end{aligned} \quad (70)$$

Note that $\mathbf{B}^{\perp}$ and $\mathbf{E}^{\parallel}$ are always continuous — they don't depend on the medium at all.

2.6 Worked Example: Capacitor and Coil

To bring together everything from this chapter, consider two simple circuits — a battery-driven capacitor and a DC-driven coil — first in vacuum, then with a dielectric or diamagnetic material inserted.

In-class exercise

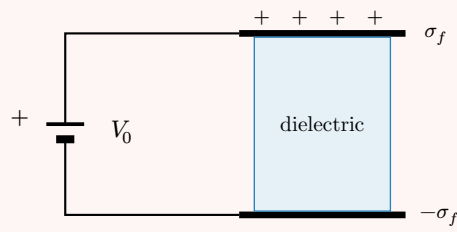


Figure 9: A battery charges a parallel-plate capacitor with a dielectric slab (e.g. glass) between the plates.

1. **Vacuum case.** Ignore the dielectric slab in Figure 9 for now. How would you determine the electric displacement $\mathbf{D}$ between the plates? Which formula/equation would you use? How are $\mathbf{D}$ and $\mathbf{E}$ connected in this case?
2. **Adding a dielectric.** Now insert a dielectric — e.g. a glass cell — between the plates, as shown in Figure 9. What happens to the glass at the microscopic level? How would you determine the polarization $\mathbf{P}$ and the field $\mathbf{E}$ in this case?

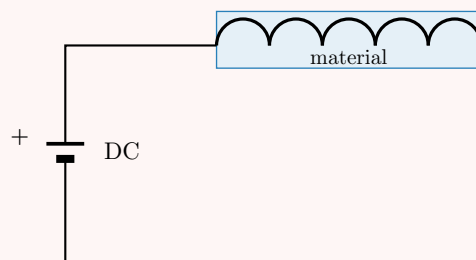


Figure 10: A DC source drives a current through a coil with a material core (e.g. glass).

3. **Vacuum case.** Ignore the material core in Figure 10 for now. Consider the coil connected to a DC source. After the current has been ramped up and is constant, a magnetic field has been generated. Which field is it exactly, and from which equation would you determine it?
4. **Adding a material core.** Now insert a glass cell inside the coil, as shown in Figure 10. What happens to the glass at the microscopic level?
5. **DC $\rightarrow$ AC.** What happens when we change the power source from DC to AC?

Qualitative answers

1. **Capacitor in vacuum.** $\mathbf{D}$ is fixed directly by the **free** charge on the plates via Gauss's law for $\mathbf{D}$, (i) in Eq. (46) — exactly the pillbox argument from Section 2.5. For plates much larger than their separation, symmetry gives a uniform $\mathbf{D}$ perpendicular to the plates, so a pillbox straddling one plate (Eq. (57)) gives $D = \sigma_f$. In vacuum, $\mathbf{D} = \epsilon_0 \mathbf{E}$, so $\mathbf{D}$ and $\mathbf{E}$ are really the same field, just in different units.

2. Capacitor with a dielectric. In the field, the bound electrons and nuclei in the glass shift slightly relative to each other, producing small induced dipole moments $\mathbf{p} = \alpha \mathbf{E}$ (the same polarizability relation from the start of this chapter) that add up to a macroscopic polarization $\mathbf{P}$, aligned with $\mathbf{E}$. As in Figure 3, the aligned dipoles cancel in the bulk of the glass but leave behind an uncompensated bound surface charge $\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}}$: negative on the face touching the + plate and positive on the face touching the – plate; opposite in sign to the free charge on that same adjacent plate, since each dipole's own – end faces the + plate (and vice versa). As a result, the free charge on the plates is partially screened reducing $\mathbf{E}$ inside the dielectric. Crucially, the pillbox argument for $\mathbf{D}$ is **unchanged**: it only encloses free charge (the bound charge sits on the dielectric's surface, outside the pillbox), so $D = \sigma_f$ still holds. Only $\mathbf{E} = (\mathbf{D} - \mathbf{P})/\epsilon_0$ changes — for a linear dielectric, $E = D/(\epsilon_0 \epsilon_r) = \sigma_f/(\epsilon_0 \epsilon_r)$, smaller than the vacuum value by a factor ϵ_r . *Extra: which quantity actually changes depends on the rest of the circuit — with the battery still connected (V_0 fixed), $E = V_0/d$ stays the same and the battery supplies extra free charge so that D (and σ_f) grow by ϵ_r ; with the battery disconnected (Q fixed), D stays the same and E (and V) drop by ϵ_r .*

3. Coil in vacuum. By the same logic, the field fixed directly by the **free** current is $\mathbf{H}$, via Ampère's law, (iv) in Eq. (46) (with $\partial_t \mathbf{D} = 0$ in the static case):

$$\oint_C \mathbf{H} \cdot d\mathbf{l} = I_{f, \text{enc}}. \quad (71)$$

For a long solenoid this gives $H = nI$, **independent of the core material** — the direct analogue of the pillbox argument for $\mathbf{D}$. In vacuum, $\mathbf{B} = \mu_0 \mathbf{H}$.

4. Coil with a material core. Glass has no permanent atomic magnetic moments (all electrons are paired), so — unlike the capacitor case — nothing aligns with the field. Instead, the changing flux as the current is switched on induces tiny orbital currents in the electron clouds that **oppose** it (the atomic-scale analogue of Lenz's law): glass, like most everyday materials, is weakly **diamagnetic**, with $\mathbf{M}$ antiparallel to $\mathbf{H}$ and χ_m of order -10^{-5} . As before, $\mathbf{H} = nI$ is unaffected by the core; only $\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M})$ changes, by about one part in 10^5 . The bound surface current $\mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{n}}$ on the glass surface is the magnetic analogue of the bound surface charge $\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}}$ from the capacitor.

5. From DC to AC. Once the source is AC, every field becomes time-dependent, and two new things happen. First, the changing polarization in the capacitor drives a real polarization current, $\mathbf{J}_p = \frac{\partial \mathbf{P}}{\partial t}$ (Eq. (38)), on top of the free current. Second — and more importantly — the electric and magnetic sectors, which were completely independent in the static examples above, become **coupled**: a changing $\mathbf{E}$ (through the $\partial_t \mathbf{D}$ term) generates $\mathbf{B}$, and a changing $\mathbf{B}$ generates $\mathbf{E}$ (Faraday's law). This is exactly the interplay we study next: in Chapter 3 we will see that this coupling between $\mathbf{E}$ and $\mathbf{B}$ gives rise to propagating **electromagnetic waves**.

3 Electromagnetic Waves

Literature: Griffiths Ch. 9.1 and 9.2

Last lectures we reviewed Maxwell's equations and derived the boundary conditions at the interface between two media. We will now study how electromagnetic waves arise from Maxwell's equations. Let's first review waves in general.

3.1 Waves: A Review

In-class exercise

What is a wave?

In simplest form: a **disturbance that propagates with a fixed shape at constant velocity**. This neglects, among other things:

- *absorption* (the wave amplitude will diminish),
- *dispersive effects* (different frequencies travel at different speeds).

In one dimension, a wave is a solution to the **wave equation**:

$$\frac{\partial^2 f}{\partial z^2} = \frac{1}{v^2} \frac{\partial^2 f}{\partial t^2} \quad (72)$$

where v is the velocity of the wave.

The solution is any function of the form

$$f(z, t) = g(z - vt) + h(z + vt) \quad (73)$$

In-class exercise

What is the physical interpretation of g and h ?

$g(z - vt)$ is a wave traveling to the *right*: the displacement found at z at time t is found again at $z + v\Delta t$ at the later time $t + \Delta t$, see Figure 11. Likewise, $h(z + vt)$ is a wave traveling to the *left*.

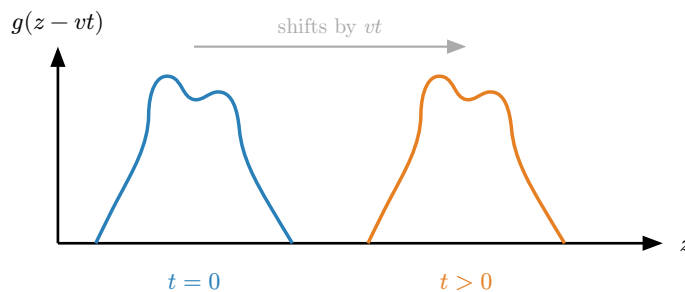


Figure 11: A wave $g(z - vt)$ traveling to the right: the same shape reappears, shifted by vt , at a later time.

3.1.1 Sinusoidal Waves

A particularly important special case is the **sinusoidal wave**:

$$f(z, t) = A \cos(k(z - vt) + \delta) \quad (74)$$

Amplitude wave number phase

To see that this is a wave: think of it at fixed t at different z , and at fixed z at different times t , then consider what happens if you change both z and t .

Terminology

- **Wavelength** $\lambda = 2\pi/k$: distance between two maxima.
- **Period** $T = 2\pi/(kv)$: in one period, a point goes through a full cycle (e.g. from $+A$ to $-A$ and back to $+A$).
- **Frequency** $\nu = 1/T = (kv)/(2\pi) = v/\lambda$: number of oscillations per unit time.
- **Angular frequency** $\omega = 2\pi\nu = kv$: number of radians per unit time.

Using $\omega = kv$, we can write the sinusoidal wave as

$$f(z, t) = A \cos(kz - \omega t + \delta) \quad (75)$$

We can use Euler's formula, $e^{i\theta} = \cos \theta + i \sin \theta$, to write a wave in complex notation:

$$f(z, t) = \text{Re}(Ae^{i(kz - \omega t + \delta)}) \quad (76)$$

This notation is very useful for solving the wave equation.

The wave equation Eq. (72) is *linear*, so that the sum of two solutions is also a solution. This is the **superposition principle**, and it is essential for combining waves (e.g. via Fourier analysis).

3.2 Waves in 3D

In 3D, there are different possibilities for the direction of oscillation. If we take the z -direction as the direction of propagation, we can have:

1. **Longitudinal waves**: oscillation in the z -direction. Example: sound waves, as density fluctuations in a medium, see Figure 12.
2. **Transverse waves**: oscillations in directions orthogonal to z .

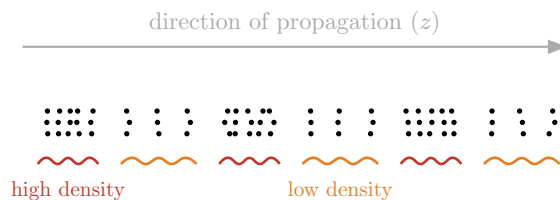


Figure 12: A longitudinal wave: regions of high and low density (compressions and rarefactions) propagate along the direction of the wave.

3.2.1 Polarization

Transverse waves in 3D have **two independent states of polarization**, because there are two directions orthogonal to the propagation direction.

The **polarization vector** $\hat{n}$ (with $\hat{n} \cdot \hat{n} = 1$) denotes the plane of oscillation, and $\hat{n} \cdot \hat{z} = 0$ for transverse waves. Figure 13 shows two examples for fixed time t , with $\hat{n} = \hat{x}$ and $\hat{n} = \hat{y}$ respectively.

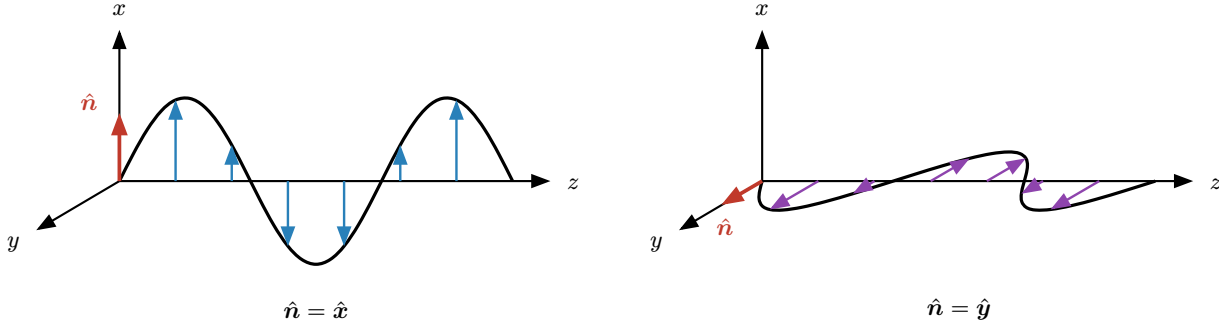


Figure 13: A transverse wave for two linear polarizations: oscillation along $\hat{x}$ (left) and along $\hat{y}$ (right). Both are orthogonal to the direction of propagation $\hat{z}$.

In general, for **linear polarization**:

$$\hat{n} = \cos \theta \hat{x} + \sin \theta \hat{y} \quad (77)$$

with **polarization angle** θ .

Circular polarization

$\hat{n}$ can also depend on time. If it rotates clockwise around the $\hat{z}$ axis (as seen looking toward the source), we speak of **right circular polarization**; if it rotates counter-clockwise, of **left circular polarization**, see [the slides for an example animation](#).

3.3 Electromagnetic Waves

Recall Maxwell's equations in vacuum, Eq. (22):

$$\begin{aligned} \nabla \cdot \mathbf{E} &= \frac{\rho}{\epsilon_0} & (i) \\ \nabla \cdot \mathbf{B} &= 0 & (ii) \\ \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} & (iii) \\ \nabla \times \mathbf{B} &= \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} & (iv) \end{aligned} \quad (78)$$

What are the consequences of these equations?

Suppose that you switch on a current $\mathbf{J}(t)$:

- $\mathbf{J}$ will vary in time,
- $\Rightarrow$ generates a $\mathbf{B}$ field, variable in time,
- $\Rightarrow$ generates an $\mathbf{E}$ field, variable in time,
- $\Rightarrow$ will again generate $\mathbf{B}$ through (iv),
- $\Rightarrow$ perpetual interplay between $\mathbf{E}$ and $\mathbf{B}$ through (iii) and (iv).

This already feels like something is oscillating. Let's play a trick to transform Maxwell's equations — first-order differential equations — into *second-order* differential equations, with the hope that they decouple.

We do this by taking the curl of (iii):

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla \times \left(-\frac{\partial \mathbf{B}}{\partial t} \right) = -\frac{\partial}{\partial t} (\nabla \times \mathbf{B}) = -\mu_0 \frac{\partial \mathbf{J}}{\partial t} - \mu_0 \varepsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}, \quad (79)$$

where in the last step we exchanged the order of the spatial curl and temporal derivative (coordinates do not depend on time), and then used (iv).

What is the left-hand side? In Section 3.7 we derive the vector calculus identity

$$[\nabla \times (\nabla \times \mathbf{E})] = [\nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E}], \quad (80)$$

where $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$ is the **Laplacian operator**, acting on all three components of $\mathbf{E}$ separately.

Using this identity and $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$ from (i):

$$\nabla \left(\frac{\rho}{\varepsilon_0} \right) - \nabla^2 \mathbf{E} = -\mu_0 \frac{\partial \mathbf{J}}{\partial t} - \mu_0 \varepsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}. \quad (81)$$

Rearranging, with all $\mathbf{E}$ -terms on the left:

$$\underbrace{\mu_0 \varepsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} - \nabla^2 \mathbf{E}}_{\square \mathbf{E}} = -\nabla \frac{\rho}{\varepsilon_0} - \mu_0 \frac{\partial \mathbf{J}}{\partial t} \quad (82)$$

The operator $\square \equiv \mu_0 \varepsilon_0 \frac{\partial^2}{\partial t^2} - \nabla^2$ is the **d'Alembert operator** (or wave operator).

Let's do the same for Maxwell's equation (iv). Due to the high symmetry between $\mathbf{E}$ and $\mathbf{B}$ in the vacuum Maxwell equations, we suspect that we will also arrive at a wave equation:

$$\nabla \times (\nabla \times \mathbf{B}) = \nabla \left(\underbrace{\nabla \cdot \mathbf{B}}_{=0} \right) - \nabla^2 \mathbf{B} = \mu_0 \nabla \times \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial}{\partial t} \left(\underbrace{\nabla \times \mathbf{E}}_{=-\frac{\partial \mathbf{B}}{\partial t}} \right), \quad (83)$$

where we used (ii) and (iv) on the left, and (iv) then (iii) on the right. Rearranging:

$$\mu_0 \varepsilon_0 \frac{\partial^2 \mathbf{B}}{\partial t^2} - \nabla^2 \mathbf{B} = \mu_0 \nabla \times \mathbf{J} \quad (84)$$

We have transformed Maxwell's equations into decoupled second-order differential equations for regions with charges and currents. (They are not *always* decoupled — e.g. Ohm's law $\mathbf{J} = \sigma \mathbf{E}$ couples them back together, as is relevant for conductors like radio antennas or the plasma in the sun's corona.)

3.4 Electromagnetic Waves in Vacuum

Let's consider $\mathbf{E}$ and $\mathbf{B}$ fields generated in some region (e.g. by an antenna). How will the fields propagate in vacuum (free space), i.e. away from the region with charges and currents?

There, Eq. (82) and Eq. (84) become:

$$\mu_0 \varepsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} - \nabla^2 \mathbf{E} = 0, \quad \mu_0 \varepsilon_0 \frac{\partial^2 \mathbf{B}}{\partial t^2} - \nabla^2 \mathbf{B} = 0 \quad (85)$$

These are 3D wave equations! (Eq. (82) and Eq. (84), with sources on the right-hand side, are called *inhomogeneous* wave equations.)

The velocity of the wave is

$$c^2 = \frac{1}{\mu_0 \varepsilon_0} \quad (86)$$

μ_0 and ε_0 can be measured in the lab, and we find $c \approx 2.9979 \times 10^8$ m/s — the speed of light! (Check yourself that $1/\sqrt{\mu_0 \varepsilon_0}$ indeed has units of m/s.)

We have stumbled onto the nature of light: it is, in fact, a disturbance in the $\mathbf{E}$ and $\mathbf{B}$ fields. With this, Maxwell's equations unify electricity, magnetism, *and* light. This is one of the great triumphs of 19th century physics.

Note that Maxwell's equations are 8 equations (1 from (i), 1 from (ii), 3 each from (iii) and (iv)), but Eq. (85) only give us 6 (3 components each for $\mathbf{E}$ and $\mathbf{B}$). Not every solution of the wave equation is a solution of Maxwell's equations — it must fulfill some extra constraints, as we will see next.

3.4.1 Plane Wave Solutions

Consider the plane wave ansatz

$$\mathbf{E}(z, t) = \mathbf{E}_0 e^{i(kz - \omega t)}, \quad (87)$$

with constant $\mathbf{E}_0$. In vacuum, $\nabla \cdot \mathbf{E} = 0$, and since $\mathbf{E}(z, t)$ is independent of x, y :

$$\partial_x E_x + \partial_y E_y + \partial_z E_z = 0 \Rightarrow E_{0z}(ik)e^{i(kz - \omega t)} = 0, \quad (88)$$

which only holds if $E_{0z} = 0$. Hence:

EM waves are **transversal**: $\mathbf{E}$ and $\mathbf{B}$ are perpendicular to the direction of propagation.

Take also $\mathbf{B}(z, t) = \mathbf{B}_0 e^{i(kz - \omega t)}$ as a plane wave. From Faraday's law, $\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$, one finds (check this yourself!) the compact relation

$$\mathbf{B}_0 = \frac{k}{\omega}(\hat{z} \times \mathbf{E}_0) \quad (89)$$

This tells us:

- $\mathbf{E}$ and $\mathbf{B}$ are both perpendicular to each other (and to $\hat{z}$), and must both be non-zero.
- $\mathbf{E}$ and $\mathbf{B}$ are *in phase* — they reach their maximum at the same time, since there is no extra phase factor in Eq. (89).
- There are only *electromagnetic* waves (not waves of $\mathbf{E}$ or $\mathbf{B}$ alone).

3.4.2 General Monochromatic Plane Waves

With $\hat{\mathbf{k}}$ the unit wave vector in the direction of propagation, and the dispersion relation $k = \omega/c$, we have for monochromatic plane waves:

$$\begin{aligned} \mathbf{E}(\mathbf{r}, t) &= E_0 e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} \hat{\mathbf{n}}, \\ \mathbf{B}(\mathbf{r}, t) &= \frac{1}{c} E_0 e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} (\hat{\mathbf{k}} \times \hat{\mathbf{n}}) = \frac{1}{c} (\hat{\mathbf{k}} \times \mathbf{E}) \end{aligned} \quad (90)$$

with $\hat{\mathbf{n}} \cdot \hat{\mathbf{k}} = 0$.

3.5 Energy and Momentum in EM Waves

The energy per unit volume stored in the fields is

$$u = \frac{1}{2} \left(\varepsilon_0 E^2 + \frac{1}{\mu_0} B^2 \right) \quad (91)$$

(check for yourself through dimensional analysis that the units work out).

From Eq. (89) we see that $B^2 = E^2/c^2$, so that

$$u = \varepsilon_0 E^2 = \frac{1}{\mu_0} B^2 \quad (92)$$

— $\mathbf{E}$ and $\mathbf{B}$ contribute equally to u — and

$$u = \varepsilon_0 E_0^2 \cos^2(kz - \omega t + \delta). \quad (93)$$

As the wave propagates, it transports energy. The **energy flux density** (amount of energy transported through a surface per unit time) is given by the Poynting vector:

$$\mathbf{S} = \frac{1}{\mu_0} (\mathbf{E} \times \mathbf{B}) \quad (94)$$

$\mathbf{S}$ points in the direction of propagation, with $|\mathbf{S}| = cu$.

The momentum density is

$$\mathbf{g} = \frac{1}{c^2} \mathbf{S}, \quad |g| = \frac{u}{c}. \quad (95)$$

3.5.1 Time-Averaged Quantities

Scale of visible light

Visible light has a wavelength between $\sim 10^2$ and 10^3 nm, which corresponds to a frequency $\nu \sim 10^{15}$ Hz, i.e. a period $T \sim 10^{-15}$ s: very fast oscillations. Usually, we measure for durations much longer than that.

We therefore consider the **time-averaged energy density** over one period $T = 2\pi/\omega$:

$$\langle u \rangle = \varepsilon_0 \frac{E_0^2}{T} \int_0^T \cos^2(kz - \omega t + \delta) dt \quad (96)$$

(we will look at time averages like this in the tutorials).

Since the time average of $\cos^2$ over a full period is $1/2$, we find the result

$$\langle u \rangle = \frac{1}{2} \varepsilon_0 E_0^2 \quad (97)$$

This immediately gives the time averages for $\mathbf{S}$ and $\mathbf{g}$ as well.

A historical puzzle

Note that the average energy density is *independent of frequency*. This led to an experimental puzzle — the photoelectric effect — which could not be explained classically, and required quantum mechanics to describe.

3.6 Electromagnetic Waves in Matter

In-class exercise

What are the wave equations for $\mathbf{E}$ and $\mathbf{B}$ in a linear and homogeneous medium, without free currents or charges? Do EM waves propagate slower than in vacuum?

In this case, the macroscopic Maxwell's equations, Eq. (46), read:

$$\begin{aligned} \text{(i)} \quad \nabla \cdot \mathbf{D} &= 0, & \text{(iii)} \quad \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t}, \\ \text{(ii)} \quad \nabla \cdot \mathbf{B} &= 0, & \text{(iv)} \quad \nabla \times \mathbf{H} &= \frac{\partial \mathbf{D}}{\partial t}. \end{aligned} \quad (98)$$

For **linear media** (Eq. (48)), $\mathbf{D} = \varepsilon \mathbf{E}$ and $\mathbf{H} = \frac{1}{\mu} \mathbf{B}$; for **homogeneous** media, ε and μ are constant (they don't vary from point to point). Plugging in, we find:

$$\begin{aligned} \text{(i)} \quad \nabla \cdot \mathbf{E} &= 0, & \text{(iii)} \quad \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t}, \\ \text{(ii)} \quad \nabla \cdot \mathbf{B} &= 0, & \text{(iv)} \quad \nabla \times \mathbf{B} &= \mu \varepsilon \frac{\partial \mathbf{E}}{\partial t}. \end{aligned} \quad (99)$$

The only difference to the vacuum case, Eq. (22), is that $\varepsilon_0 \rightarrow \varepsilon$ and $\mu_0 \rightarrow \mu$. Thus, we find EM waves that propagate at speed

$$v = \frac{1}{\sqrt{\varepsilon \mu}} = \frac{c}{n} \quad (100)$$

where

$$n \equiv \sqrt{\frac{\varepsilon \mu}{\varepsilon_0 \mu_0}} \quad (101)$$

is the **index of refraction**.

All previous results carry over to the case of a medium, with $\varepsilon_0 \rightarrow \varepsilon$ and $\mu_0 \rightarrow \mu$. For most materials, $\mu \approx \mu_0$, so that

$$n \approx \sqrt{\varepsilon_r} = \sqrt{\varepsilon/\varepsilon_0}, \quad (102)$$

where ε_r is called the **dielectric constant**, which might depend on frequency. Almost always $\varepsilon_r > 1$, so waves travel *slower* in media.

The energy density and Poynting vector become

$$u = \frac{1}{2} \left(\varepsilon E^2 + \frac{1}{\mu} B^2 \right), \quad \mathbf{S} = \frac{1}{\mu} (\mathbf{E} \times \mathbf{B}). \quad (103)$$

Why does light slow down in matter?

Mathematically, replacing $\varepsilon_0 \rightarrow \varepsilon$ and $\mu_0 \rightarrow \mu$ is a pretty trivial result — but physically, it is quite astonishing. As waves pass through matter, they polarize and magnetize the molecules, producing oscillating dipoles that create their own fields. These new fields combine with the original wave to give a wave at the *same* frequency, but a *different* speed.

This is a non-trivial consequence of linearity, and is the reason materials like glass or water are transparent: the wave that emerges is still a single coherent wave at the original frequency, just slowed down.

3.7 Deriving the curl of a curl identity

The “BAC – CAB” rule for the curl of a curl

Similarly to the vector identity

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B}) \quad (104)$$

(“BAC – CAB”), one can show an analogous identity for the differential operator ∇ :

$$\nabla \times (\nabla \times \mathbf{C}) = \nabla(\nabla \cdot \mathbf{C}) - \nabla^2 \mathbf{C}, \quad (105)$$

where ∇^2 is the **Laplacian operator**,

$$\nabla^2 f = \partial_x^2 f + \partial_y^2 f + \partial_z^2 f = \sum_i \partial_i^2 f \equiv \partial_i \partial_i f, \quad (106)$$

acting on each component of $\mathbf{C}$ separately.

Proof (in Einstein notation): look at the i -th component,

$$[\mathbf{A} \times (\mathbf{B} \times \mathbf{C})]_i = \varepsilon_{ijk} A_j (\mathbf{B} \times \mathbf{C})_k = \varepsilon_{ijk} A_j \varepsilon_{klm} B_l C_m. \quad (107)$$

Using the cyclic property $\varepsilon_{ijk} = \varepsilon_{kij}$ and the identity $\varepsilon_{kij} \varepsilon_{klm} = \delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}$ (which follows from the definition of ε_{ijk} , in the same way as in the note on the Levi-Civita tensor in Chapter 1), this becomes

$$\begin{aligned} \varepsilon_{ijk} A_j \varepsilon_{klm} B_l C_m &= (\delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}) A_j B_l C_m \\ &= B_i (A_j C_j) - C_i (A_j B_j) = [\mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})]_i. \end{aligned} \quad (108)$$

The same computation with $\mathbf{A} = \mathbf{B} = \nabla$ (so that $A_j \rightarrow \partial_j$) and $\mathbf{C} \rightarrow \mathbf{C}$ gives

$$\begin{aligned}
[\nabla \times (\nabla \times \mathbf{C})]_i &= \varepsilon_{ijk} \partial_j \varepsilon_{klm} \partial_l C_m \\
&= \partial_i (\partial_j C_j) - \partial_j \partial_j C_i = [\nabla (\nabla \cdot \mathbf{C}) - \nabla^2 \mathbf{C}]_i. \checkmark
\end{aligned} \tag{109}$$

4 Reflection and Transmission

Literature: Griffiths Ch. 9.3

Last lecture we saw that Maxwell's equations lead to EM waves. Today we continue our discussion of EM waves in matter. In particular, we want to see what happens when EM waves traverse the boundary between two media, and use the boundary conditions we derived previously (Section 2.5).

4.1 Normal Incidence

In-class exercise

What happens when a wave passes an interface, e.g. air $\rightarrow$ water or water $\rightarrow$ air? (Think of your personal experience!)

For an incident wave, there will be a **reflected** and a **transmitted** wave, both for $\mathbf{E}$ and $\mathbf{B}$, which we call $\mathbf{E}_I, \mathbf{E}_R, \mathbf{E}_T$ and $\mathbf{B}_I, \mathbf{B}_R, \mathbf{B}_T$ respectively. The reflected and transmitted fields are determined from the boundary conditions we already derived (Section 2.5).

Let's first look at the case of **normal incidence**: the incident $\mathbf{E}$ field is a plane wave traveling in the $+z$ direction, polarized along $\hat{\mathbf{x}}$, with the boundary in the xy plane at $z = 0$, see Figure 14.

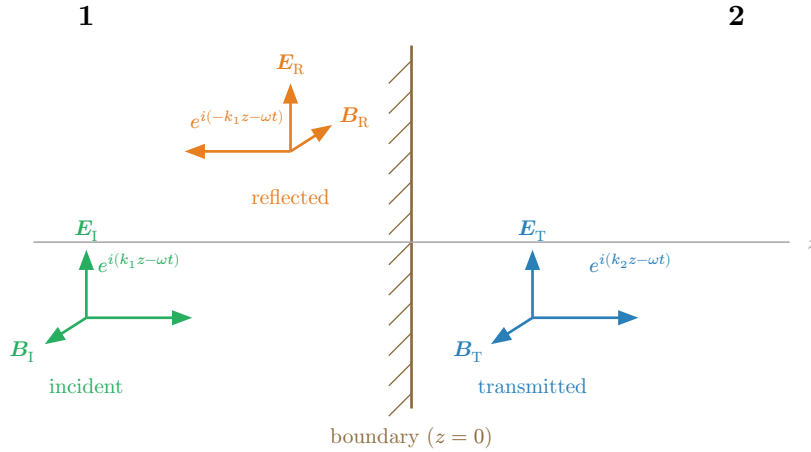


Figure 14: Normal incidence: the incident wave (medium 1) generates a reflected wave (medium 1) and a transmitted wave (medium 2) at the boundary $z = 0$.

The incoming wave is

$$\mathbf{E}_I(z, t) = E_{0I} e^{i(k_1 z - \omega t)} \hat{\mathbf{x}}, \quad \mathbf{B}_I(z, t) = B_{0I} e^{i(k_1 z - \omega t)} \hat{\mathbf{y}}, \quad (110)$$

where, as we saw last lecture, $\mathbf{B} = \frac{1}{v} (\hat{\mathbf{k}} \times \mathbf{E})$, so that

$$B_{0I} = \frac{E_{0I}}{v_1} \quad (111)$$

↑
wave velocity in medium (1)

with

$$v_i = \frac{1}{\sqrt{\epsilon_i \mu_i}} = \frac{c}{n_i}, \quad n_i > 1, \quad i = 1, 2. \quad (112)$$

A **reflected** wave is generated,

$$\mathbf{E}_R(z, t) = E_{0R} e^{i(-k_1 z - \omega t)} \hat{\mathbf{x}}, \quad \mathbf{B}_R(z, t) = -\frac{E_{0R}}{v_1} e^{i(-k_1 z - \omega t)} \hat{\mathbf{y}}, \quad (113)$$



required so $\mathbf{S}$ points along propagation direction $-z$

And a **transmitted** wave is generated,

$$\mathbf{E}_T(z, t) = E_{0T} e^{i(k_2 z - \omega t)} \hat{\mathbf{x}}, \quad \mathbf{B}_T(z, t) = \frac{E_{0T}}{v_2} e^{i(k_2 z - \omega t)} \hat{\mathbf{y}}. \quad (114)$$

We show the incident, reflected, and transmitted waves in Figure 14.

Now, given E_{0I} , we would like to determine E_{0R} and E_{0T} . For this we use the boundary conditions (Section 2.5 and a [recap on the slides](#)), noting that the boundary is a surface in the xy plane at $z = 0$. We assume linear media without surface charges or currents.

4.1.1 Boundary Conditions and Amplitudes

In general, the field on the left, $\mathbf{E}_I + \mathbf{E}_R$ and $\mathbf{B}_I + \mathbf{B}_R$, must join with the fields on the right, $\mathbf{E}_T$ and $\mathbf{B}_T$, so that the boundary conditions are satisfied. The fields have no components perpendicular to the surface (no z component), so boundary conditions for the perpendicular components, (i) and (ii) in Eq. (70), are trivial.

From (iii), stating that $\mathbf{E}^\parallel$ is continuous across the boundary, we have:

$$\underbrace{\mathbf{E}_R + \mathbf{E}_I}_{\mathbf{E}_1^\parallel} = \underbrace{\mathbf{E}_T}_{\mathbf{E}_2^\parallel} \quad (115)$$

and since the boundary is at $z = 0$ we find

$$\Rightarrow E_{0R} + E_{0I} = E_{0T}, \quad (116)$$

where we have divided by $e^{-i\omega t}$.

With boundary condition (iv) gives

$$\underbrace{\frac{1}{\mu_1} \left(-\frac{E_{0R}}{v_1} + \frac{E_{0I}}{v_1} \right)}_{B_1^\parallel / \mu_1} = \underbrace{\frac{E_{0T}}{\mu_2 v_2}}_{B_2^\parallel / \mu_2} \quad (117)$$

$$\Rightarrow E_{0I} - E_{0R} = \underbrace{\frac{\mu_1 v_1}{\mu_2 v_2}}_{\beta = \frac{\mu_1 n_2}{\mu_2 n_1}} E_{0T} \quad (118)$$

We have two unknowns, E_{0R} and E_{0T} , and two equations, Eq. (116) and Eq. (118), which we can solve, e.g. by solving Eq. (118) for E_{0T} and plugging this into Eq. (116):

$$E_{0R} + E_{0I} = \frac{1}{\beta} (E_{0I} - E_{0R}). \quad (119)$$

Solving for E_{0R} we find

$$E_{0R} = \frac{1 - \beta}{1 + \beta} E_{0I} \quad (120)$$

and plugging this back into Eq. (118) we get

$$E_{0T} = \frac{2}{1 + \beta} E_{0I} \quad (121)$$

with

$$\beta \equiv \frac{\mu_1 v_1}{\mu_2 v_2} = \frac{\mu_1 n_2}{\mu_2 n_1} \quad (122)$$

The special case $\mu_1 \approx \mu_2 \approx \mu_0$

As mentioned before, for most media $\mu \approx \mu_0$. Under this assumption, $\beta = v_1/v_2 = n_2/n_1$, and Eq. (120) and Eq. (121) become

$$E_{0R} = \frac{n_1 - n_2}{n_1 + n_2} E_{0I}, \quad E_{0T} = \frac{2n_1}{n_1 + n_2} E_{0I}. \quad (123)$$

(Check this yourself!)

4.1.2 Reflection and Transmission Coefficients

What fraction of the incident energy is transmitted, and what fraction is reflected? For this we look at the incoming intensity, i.e. the time-averaged Poynting vector (cf. Tutorial 3) for a plane wave in a linear medium:

$$I = \underbrace{\langle \mathbf{S} \rangle}_{\text{time avg. over one cycle}} = \underbrace{\frac{1}{2} \varepsilon v E_0^2}_{\text{for plane waves in linear media}} \quad (124)$$

With this, we can determine the reflection and transmission coefficients from Eq. (123):

$$R = \frac{I_R}{I_I} = \left(\frac{n_1 - n_2}{n_1 + n_2} \right)^2, \quad (125)$$

where we have used that the reflected and incident waves are in the same medium, so v_1 and ε_1 cancel. And

$$T = \frac{I_T}{I_I} = \frac{\varepsilon_2}{\varepsilon_1} \frac{v_2}{v_1} \left(\frac{E_{0T}}{E_{0I}} \right)^2 = \frac{4n_1 n_2}{(n_1 + n_2)^2} \quad (126)$$

$\frac{n_2^2}{n_1^2}$ since $n_i \sim \sqrt{\varepsilon_i/\varepsilon_0}$ $\xrightarrow{\text{Eq. (123)}}$

And it's easy to verify that

$$R + T = 1, \quad (127)$$

as required from energy conservation. (Check this yourself!)

Connection to quantum mechanics

We will revisit reflection and transmission coefficients in quantum mechanics, when discussing phenomena like scattering and tunneling.

In-class exercise

Discuss with your neighbor: Is the derivation clear? What happens when $n_1 = n_2$? What happens when $n_1 \gg n_2$?

4.2 Reflection and Transmission at Oblique Incidence

Let's turn to the more general case of **oblique incidence**, i.e. where light is not coming in perpendicular to the boundary, but at an angle θ_I (think e.g. of light shining through a window). Normal incidence is just the special case $\theta_I = 0$.

We have the following situation, see Figure 15:

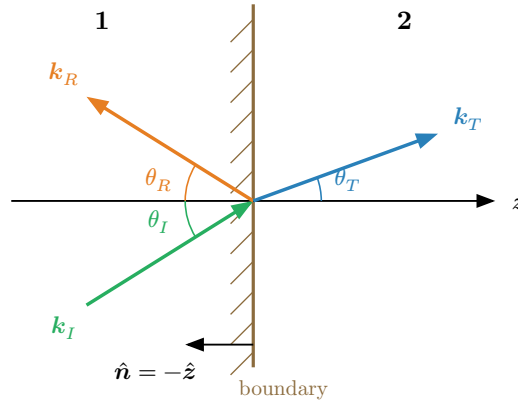


Figure 15: Oblique incidence: the incident, reflected, and transmitted wave vectors, together with the angles they make with the surface normal $\hat{n} = -\hat{z}$.

where the incoming wave is now

$$\mathbf{E}_I = \mathbf{E}_{0I} e^{i(\mathbf{k}_I \cdot \mathbf{r} - \omega t)}, \quad \mathbf{B}_I = \frac{1}{v_1} (\hat{\mathbf{k}}_I \times \mathbf{E}_I). \quad (128)$$

(Remember: the physical fields are the real part of these complex expressions and also note that we're still assuming monochromatic waves.)

Similarly, the transmitted and reflected waves are

$$\mathbf{E}_R = \mathbf{E}_{0R} e^{i(\mathbf{k}_R \cdot \mathbf{r} - \omega t)}, \quad \mathbf{B}_R = \frac{1}{v_1} (\hat{\mathbf{k}}_R \times \mathbf{E}_R), \quad (129)$$

$$\mathbf{E}_T = \mathbf{E}_{0T} e^{i(\mathbf{k}_T \cdot \mathbf{r} - \omega t)}, \quad \mathbf{B}_T = \frac{1}{v_2} (\hat{\mathbf{k}}_T \times \mathbf{E}_T). \quad (130)$$

All waves have the same frequency ω , determined e.g. by the light source. Remembering the **dispersion relation** for plane waves, $\omega = kv$, we find in our case:

$$\omega = k_I v_1 = k_R v_1 = k_T v_2, \quad (131)$$

or, in terms of just the magnitude of the wave vectors,

$$k_I = k_R = \frac{v_2}{v_1} k_T = \frac{n_1}{n_2} k_T. \quad (132)$$

4.2.1 Laws of Geometric Optics

We again use the boundary conditions, where

$$\mathbf{E}_1 = \mathbf{E}_I + \mathbf{E}_R, \quad \mathbf{B}_1 = \mathbf{B}_I + \mathbf{B}_R, \quad \mathbf{E}_2 = \mathbf{E}_T, \quad \mathbf{B}_2 = \mathbf{B}_T. \quad (133)$$

Parallel and perpendicular components

We can get the parallel ($\parallel$) and perpendicular ($\perp$) components with the help of the surface normal $\hat{\mathbf{n}}$, e.g.

$$\mathbf{E}_1^\perp = \mathbf{E}_1 \cdot \hat{\mathbf{n}}, \quad \mathbf{E}_1^\parallel = \mathbf{E}_1 - (\mathbf{E}_1 \cdot \hat{\mathbf{n}})\hat{\mathbf{n}}. \quad (134)$$

From boundary condition (i) at $z = 0$, we have

$$\varepsilon_1 \left(\mathbf{E}_{0I}^\parallel e^{i(\mathbf{k}_I \cdot \mathbf{r} - \omega t)} + \mathbf{E}_{0R}^\parallel e^{i(\mathbf{k}_R \cdot \mathbf{r} - \omega t)} \right) = \varepsilon_2 \mathbf{E}_{0T}^\parallel e^{i(\mathbf{k}_T \cdot \mathbf{r} - \omega t)}. \quad (135)$$

With $\mathbf{r} = (x, y, 0)$. The x, y, t dependence is only in the exponents, so for any point on the boundary and any time t we must have

$$\mathbf{k}_I \cdot \mathbf{r} = \mathbf{k}_R \cdot \mathbf{r} = \mathbf{k}_T \cdot \mathbf{r}, \quad (136)$$

(simply divide both sides by the exponential factors $e^{-i\omega t}$ in Eq. (135) to see this). Multiplying out the dot products, we find

$$k_I^x x + k_I^y y = k_R^x x + k_R^y y = k_T^x x + k_T^y y, \quad (137)$$

which only holds if the components are separately equal — e.g. for $x = 0$: $k_I^y = k_R^y = k_T^y$ or $y = 0$:

$$k_I^x = k_R^x = k_T^x. \quad (138)$$

Let's choose a coordinate system where $\mathbf{k}_I$ lies in the xz plane like in Figure 15 ($k_I^y = 0$); then Eq. (138) forces $\mathbf{k}_R$ and $\mathbf{k}_T$ to also lie in the xz plane. From this we find the **fundamental laws of geometric optics**:

1. **First Law:** The incident, reflected, and transmitted wave vectors form a plane — the *plane of incidence* — that also contains the normal vector $\hat{\mathbf{n}}$ of the interface.
2. **Second Law:** $\theta_I = \theta_R$: the angle of incidence equals the angle of reflection. This follows from Eq. (138),

$$k_I^x = k_I \sin \theta_I = k_R \sin \theta_R = k_R^x, \quad \text{and} \quad k_I = k_R. \quad (139)$$

3. **Third Law (Snell's law):**

$$\frac{\sin(\theta_T)}{\sin(\theta_I)} = \frac{n_1}{n_2}, \quad (140)$$

which again follows from Eq. (138), $k_I \sin \theta_I = k_T \sin \theta_T$, together with $k_{I,T} = \omega/v_{1,2}$.

In-class exercise

Discuss with your neighbor: Is the derivation clear?

Now that we've taken care of the exponential factors in the boundary conditions, we can continue with finding the amplitudes. The exponential factors cancel, and we're left with:

$$\begin{aligned}
 \text{(i)} \quad & \varepsilon_1(\mathbf{E}_{0I} + \mathbf{E}_{0R})^\perp = \varepsilon_2(\mathbf{E}_{0T})^\perp \\
 \text{(ii)} \quad & (\mathbf{B}_{0I} + \mathbf{B}_{0R})^\perp = (\mathbf{B}_{0T})^\perp \\
 \text{(iii)} \quad & (\mathbf{E}_{0I} + \mathbf{E}_{0R})^\parallel = (\mathbf{E}_{0T})^\parallel \\
 \text{(iv)} \quad & \frac{1}{\mu_1}(\mathbf{B}_{0I} + \mathbf{B}_{0R})^\parallel = \frac{1}{\mu_2}(\mathbf{B}_{0T})^\parallel
 \end{aligned} \tag{141}$$

In our coordinate system, the perpendicular component ($\perp$) is along $\hat{\mathbf{z}}$. Note that Eqs. (iii) and (iv) contain two equations for both the x and y components separately.

4.2.2 Amplitudes: Polarization in the Plane of Incidence

Let's assume that the polarization of the incident wave is in the plane of incidence, see Figure 16.

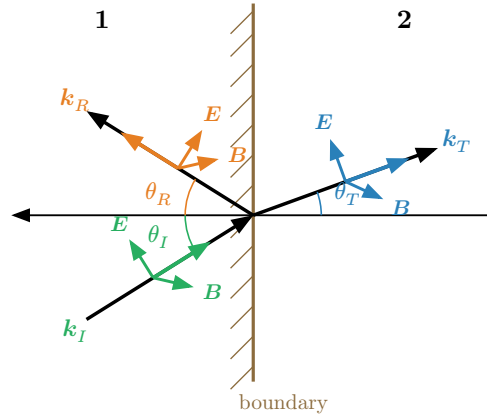


Figure 16: Oblique incidence with polarization in the plane of incidence: $\mathbf{E}$ lies in the plane of the page for all three waves, while $\mathbf{B}$ points out of the page.

Then (i), the perpendicular component, becomes

$$\varepsilon_1(-E_{0I} \sin \theta_I + E_{0R} \sin \theta_R) = \varepsilon_2(-E_{0T} \sin \theta_T), \tag{142}$$

using $\theta_I = \theta_R$ (law 2):

$$\begin{aligned}
 \varepsilon_1 \sin \theta_I (-E_{0I} + E_{0R}) &= -\varepsilon_2 \sin \theta_T E_{0T} \\
 \Leftrightarrow E_{0R} - E_{0I} &= -\frac{\varepsilon_2 \sin(\theta_T)}{\varepsilon_1 \sin(\theta_I)} E_{0T}.
 \end{aligned} \tag{143}$$

Using Snell's law (Eq. (140)) and multiply with -1 to flip the sign, this becomes

$$\begin{aligned}
E_{0I} - E_{0R} &= \frac{\varepsilon_2}{\varepsilon_1} \frac{n_1}{n_2} E_{0T} = \frac{\mu_1 n_2}{\underbrace{\mu_2 n_1}_{\equiv \beta}} E_{0T} \\
&= \beta E_{0T}
\end{aligned} \tag{144}$$

— the same β as in Eq. (122) for which we used $v = 1/\sqrt{\varepsilon\mu} \Leftrightarrow \varepsilon = 1/(\mu v^2) = n^2/(\mu c^2)$.

Boundary condition (ii) does not add anything, since $\mathbf{B}$ has no z component.

Boundary condition (iii), the parallel component, becomes

$$\begin{aligned}
E_{0I} \cos \theta_I + E_{0R} \cos \theta_R &= E_{0T} \cos \theta_T \\
\Leftrightarrow E_{0I} + E_{0R} &= \frac{\cos(\theta_T)}{\underbrace{\cos(\theta_I)}_{\equiv \alpha}} E_{0T} \\
&= \alpha E_{0T}
\end{aligned} \tag{145}$$

using law 2 again.

Boundary condition (iv) gives the same result as (i) by using $\mathbf{B} = \frac{1}{v}(\hat{\mathbf{k}} \times \mathbf{E})$ (check this yourself!).

With Eq. (144) and Eq. (145), we again have two equations for two unknown amplitudes E_{0R} and E_{0T} .

Plugging in Eq. (145) into Eq. (144) gives

$$\begin{aligned}
E_{0I} - E_{0R} &= \frac{\beta}{\alpha} (E_{0I} + E_{0R}) \\
\Leftrightarrow E_{0R} &= \frac{1 - \frac{\beta}{\alpha}}{1 + \frac{\beta}{\alpha}} E_{0I} \\
\Leftrightarrow E_{0R} &= \frac{\alpha - \beta}{\alpha + \frac{\beta}{\alpha}} E_{0I}.
\end{aligned} \tag{146}$$

Plugging this back into Eq. (145) we find

$$E_{0R} = \frac{\alpha - \beta}{\alpha + \beta} E_{0I}, \quad E_{0T} = \frac{2}{\alpha + \beta} E_{0I} \tag{147}$$

with

$$\alpha \equiv \frac{\cos(\theta_T)}{\cos(\theta_I)}, \quad \beta \equiv \frac{\mu_1 n_2}{\mu_2 n_1} \tag{148}$$

These are the **Fresnel equations** for the case of polarization in the plane of incidence.

Using Snell's law (Eq. (140)) we can show that α and thus the amplitudes of the reflected and transmitted waves depend only on the angle of incidence θ_I and the refractive indices n_1 and n_2 of the two media:

$$\alpha = \frac{\sqrt{1 - \sin^2 \theta_T}}{\cos \theta_I} = \frac{\sqrt{1 - \left(\frac{n_1}{n_2}\right)^2 \sin^2 \theta_I}}{\cos \theta_I}. \quad (149)$$

Consistency checks

- For $\theta_I = 0$, $\alpha = 1$, and we recover normal incidence, Eq. (120) and Eq. (121).
- For $\theta_I = 90^\circ$ (grazing incidence), $\cos \theta_I \rightarrow 0$ so $\alpha \rightarrow \infty$, and the wave is totally reflected.

4.2.3 The Brewster Angle

There is also an intermediate angle, known as the **Brewster angle** θ_B , where E_{0R} vanishes. This occurs when θ_I is such that $\alpha = \beta$. Expressing α in terms of θ_I using Eq. (149) and $\cos^2 \theta_I = 1 - \sin^2 \theta_I$, one finds

$$\sin^2 \theta_B = \frac{1 - \beta^2}{(n_1/n_2)^2 - \beta^2}. \quad (150)$$

For $\mu_1 \approx \mu_2 \approx \mu_0$ we have $\beta = \frac{n_2}{n_1}$, and this simplifies to

$$\sin^2 \theta_B = \frac{\beta^2}{1 + \beta^2}. \quad (151)$$

Using trigonometric identities, we can also express this in terms of the tangent:

$$\tan \theta_B = \frac{n_2}{n_1}. \quad (152)$$

This only occurs for polarization in the plane of incidence — this is why polarized (sun)glasses reduce the glare off a horizontal surface: they block the (unpolarized) light component reflected at close to the Brewster angle.

4.2.4 Reflection and Transmission Coefficients (Oblique Incidence)

Lastly, we derive the transmission and reflection coefficients for the general case of oblique incidence. The intensity (power per unit area) of the incident wave hitting the interface is $I = \mathbf{S} \cdot \hat{\mathbf{z}}$, so that

$$I_I = \frac{1}{2} \varepsilon_1 v_1 E_{0I}^2 \cos \theta_I, \quad (153)$$

while the reflected and transmitted intensities are

$$I_R = \frac{1}{2} \varepsilon_1 v_1 E_{0R}^2 \cos \theta_R, \quad I_T = \frac{1}{2} \varepsilon_2 v_2 E_{0T}^2 \cos \theta_T. \quad (154)$$

This gives

$$\begin{aligned} R &= \frac{I_R}{I_I} = \left(\frac{E_{0R}}{E_{0I}} \right)^2 = \left(\frac{\alpha - \beta}{\alpha + \beta} \right)^2, \\ T &= \frac{I_T}{I_I} = \frac{\varepsilon_2 v_2}{\varepsilon_1 v_1} \left(\frac{E_{0T}}{E_{0I}} \right)^2 \frac{\cos(\theta_T)}{\cos(\theta_I)} = \alpha \beta \left(\frac{2}{\alpha + \beta} \right)^2 \end{aligned} \quad (155)$$

Again, you may check that $R + T = 1$, as required by energy conservation.

Example: air to glass

Consider the interface between air ($n_1 = 1$) and glass ($n_2 = 1.5$), so that $\beta = n_2/n_1 = 1.5$. See the slides for plots of R and T as a function of θ_I .

Next lecture: we will see what happens when waves propagate in media with free charges and surface currents, such as conductors — this leads to the phenomena of absorption.

5 Absorption and Dispersion

Literature: Griffiths Ch. 9.4

Last lecture we considered the reflection and transmission of EM waves at the boundary between two **non-conducting** media, i.e. media without free surface charges. This simplified our boundary conditions for the parallel and perpendicular $\mathbf{E}$ and $\mathbf{B}$ components.

In this lecture, we'll study what happens now at the boundary of a non-conducting and a **conducting** medium.

In the second part, we will return to the propagation of EM waves in non-conducting media, but now allowing for a **frequency-dependent permittivity** $\varepsilon(\omega)$, which leads to **dispersion**.

5.1 Electromagnetic Waves in Conductors

In a conductor, the free current density is given by (the generalized form of) **Ohm's law**:

$$\mathbf{J}_f = \sigma \mathbf{E}, \quad (156)$$

where σ is called the **conductivity**.

Connection to $U = RI$

The more familiar form of Ohm's law, $U = RI$, is a special case of Eq. (156). You can retrieve it by integrating the electric field along a wire of length L and cross-sectional area A .

For a constant field $\mathbf{E}$ along the wire, the voltage is

$$U = \int_0^L \mathbf{E} \cdot d\mathbf{l} = EL \quad (157)$$

and thus $E = \frac{V}{L}$. If $\mathbf{E}$ is uniform across the cross-section, the current is

$$I = \int_A \mathbf{J}_f \cdot d\mathbf{A} = J_f A \quad (158)$$

or $J_f = \frac{I}{A}$.

Plugging everything into Eq. (156), we recover $U = RI$ with the resistance is $R = \frac{L}{\sigma A}$ and the current is $I = J_f A$.

Plugging this into Maxwell's equations (for a linear, homogeneous medium):

$$\begin{aligned} \text{(i)} \quad \nabla \cdot \mathbf{E} &= \frac{\rho_f}{\varepsilon} & \text{(iii)} \quad \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} \\ \text{(ii)} \quad \nabla \cdot \mathbf{B} &= 0 & \text{(iv)} \quad \nabla \times \mathbf{B} &= \mu\sigma \mathbf{E} + \mu\varepsilon \frac{\partial \mathbf{E}}{\partial t} \end{aligned} \quad (159)$$

You will show in the exercises that the free charge density is then a function of time and decays,

$$\rho_f(t) = \rho_f(0)e^{-\frac{\sigma}{\varepsilon}t}. \quad (160)$$


 $\equiv 1/\tau$

Thus, if you put some free charge on a conductor, it will dissipate on a time scale $\tau = \varepsilon/\sigma$, and “flow out to the edges.” If you wait sufficiently long, (i) in Eq. (159) becomes

$$\nabla \cdot \mathbf{E} = 0. \quad (161)$$

5.1.1 Wave Equation with a Complex Wavenumber

In the exercises you will also show that this set of Maxwell’s equations can be transformed into a wave equation (following the same steps as in Chapter 3), and that it is solved by plane waves with **complex wavenumbers**,

$$\tilde{k} = k + i\kappa. \quad (162)$$

The complex part leads to an **attenuation** of the wave. Writing this out explicitly — I’m now including the tilde on $\mathbf{E}$ and $\mathbf{B}$ to remind you that they are complex:

$$\tilde{\mathbf{X}}(z, t) = \tilde{\mathbf{X}}_0 e^{-\kappa z} e^{i(kz - \omega t)}, \quad \mathbf{X} = \mathbf{E} \text{ or } \mathbf{B}, \quad (163)$$

with

$$\begin{aligned} k &= \omega \sqrt{\frac{\varepsilon\mu}{2}} \left[\sqrt{1 + \left(\frac{\sigma}{\varepsilon\omega}\right)^2} + 1 \right]^{\frac{1}{2}}, \\ \kappa &= \omega \sqrt{\frac{\varepsilon\mu}{2}} \left[\sqrt{1 + \left(\frac{\sigma}{\varepsilon\omega}\right)^2} - 1 \right]^{\frac{1}{2}}. \end{aligned} \quad (164)$$

Consequence: waves only penetrate a finite amount into a conductor, characterized by the **skin depth**

$$d \equiv \frac{1}{\kappa}. \quad (165)$$

Example: skin depth of silver at optical frequencies

Metals like silver have high conductivity, $\sigma \sim 10^7 (\Omega \cdot \text{m})^{-1}$, so the $(\sigma/(\varepsilon\omega))^2$ term dominates in κ , Eq. (164):

$$\kappa \approx \omega \sqrt{\frac{\varepsilon\mu}{2}} \sqrt{\frac{\sigma}{\varepsilon\omega}} \approx \sqrt{\frac{\omega\mu_0\sigma}{2}}, \quad (166)$$

so that

$$d = \frac{1}{\kappa} \approx \sqrt{\frac{2}{\omega\mu_0\sigma}} = \sqrt{\frac{2}{10^{15} \cdot 4\pi \times 10^{-7} \cdot 10^7}} \text{ m} \quad (167)$$

$$\approx 1.3 \times 10^{-8} \text{ m} = 13 \text{ nm},$$

where we have used $\omega \approx 10^{15} \text{ Hz}$.

See also [this skin-depth calculator](#).

The real part of $\tilde{k}$ determines the wavelength, propagation speed, and index of refraction, as before:

$$\lambda = \frac{2\pi}{k}, \quad v = \frac{\omega}{k}, \quad n = \frac{ck}{\omega}. \quad (168)$$

In-class exercise

Are the waves still transverse?

5.1.2 Polarization and Phase of $\mathbf{E}$ and $\mathbf{B}$

Let's choose the electric field to be polarized along the x -axis and travelling along the z -direction:

$$\tilde{\mathbf{E}} = \tilde{E}_0 e^{-\kappa z} e^{i(kz - \omega t)} \hat{\mathbf{x}}, \quad (169)$$

and from Maxwell's equation (iii) we find the same relation as for plane waves, Eq. (90),

$$\tilde{\mathbf{B}} = \frac{\tilde{k}}{\omega} \tilde{E}_0 e^{-\kappa z} e^{i(kz - \omega t)} \hat{\mathbf{y}}. \quad (170)$$

The complex number $\tilde{k}$ in the magnetic field will lead to a **phase difference** between the $\mathbf{E}$ and $\mathbf{B}$ fields.

How large is the phase difference? Remember that we can rewrite a complex number as

$$\tilde{k} = k + i\kappa = K e^{i\phi}, \quad K = \sqrt{k^2 + \kappa^2}, \quad \tan \phi = \frac{\kappa}{k}. \quad (171)$$

Thus, the complex amplitudes of the fields are related by

$$\tilde{B}_0 = B_0 e^{i\delta_B} = \frac{K e^{i\phi}}{\omega} E_0 e^{i\delta_E} \Rightarrow \delta_B - \delta_E = \phi : \quad (172)$$

the magnetic field **lags behind** the electric field ([see the slides for a plot](#)).

5.2 Reflection off a Conductor

Now we can carry out the analysis of what happens when an EM wave hits the boundary between a non-conducting medium and a conductor. For simplicity, let's look at the case of **normal incidence**: waves polarized in the $\hat{\mathbf{x}}$ direction, propagating along the z -axis, i.e. $\theta_I = 0$.

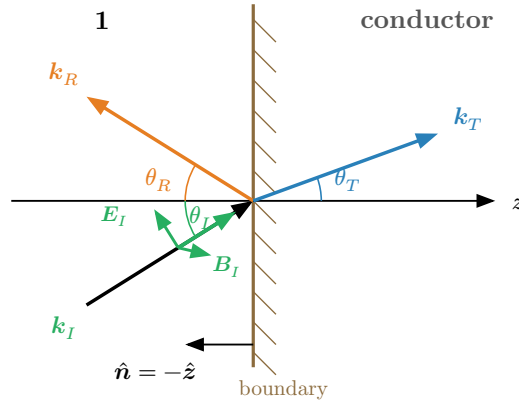


Figure 17: A wave hits the boundary between a non-conducting medium (1) and a conductor. For the derivation in the text, we specialize to normal incidence, $\theta_I = 0$.

Now, we use again the general boundary conditions, Eq. (69). (see the slides for a reminder). For ohmic conductors, there can be no free surface current $\mathbf{K}_f$, as this would require an infinite electric field at the boundary:

$$\mathbf{K}_f = 0. \quad (173)$$

With $\theta_I = 0$, we don't have a polarization perpendicular to the boundary (i.e. parallel to the surface normal $\hat{\mathbf{n}}$), so boundary condition (i) yields

$$\sigma_f = 0. \quad (174)$$

5.2.1 Parallel Components

For the parallel components, the analysis is the same as last lecture (Chapter 4). The only difference: the wave vector $\tilde{\mathbf{k}}_2$, in the conductor, is now complex. Going through the same steps as before, we get

$$\tilde{E}_{0R} = \frac{1 - \tilde{\beta}}{1 + \tilde{\beta}} \tilde{E}_{0I}, \quad \tilde{E}_{0T} = \frac{2}{1 + \tilde{\beta}} \tilde{E}_{0I}, \quad (175)$$

with

$$\tilde{\beta} = \frac{\mu_1 v_1}{\mu_2 \omega} \tilde{\mathbf{k}}_2, \quad (176)$$

$\uparrow$
 $\in \mathbb{C}$ (complex number)

so that $\tilde{\beta}$ is itself a complex number.

For a **perfect conductor**, $\sigma \rightarrow \infty$. From Eq. (164), both $k_2 \rightarrow \infty$ and $\kappa_2 \rightarrow \infty$, so $\tilde{\mathbf{k}}_2 \rightarrow \infty$ and thus $\tilde{\beta} \rightarrow \infty$. As a result, we find

$$\tilde{E}_{0R} \rightarrow -\tilde{E}_{0I}, \quad \tilde{E}_{0T} \rightarrow 0. \quad (177)$$

There's no transmitted wave — only a reflected wave, which has a 180° phase shift compared to the incoming wave. **This is how a mirror works!**

Mirrors in practice

A household mirror usually has a thin layer of silver on the back of a glass plate — the glass is just there to support the thin silver layer.

In-class exercise

Discuss with your neighbor: are these derivations clear?

5.3 Frequency Dependence of the Permittivity

Literature: Griffiths Ch. 9.4.3

5.3.1 Recap

We have seen that the propagation of EM waves in media depends on three material properties.

In-class exercise

Can you name them?

Here's a summary:

1. **Permittivity** ε and **permeability** μ , giving

$$v = \frac{1}{\sqrt{\mu\varepsilon}} = \frac{1}{n\sqrt{\varepsilon_0\mu_0}} = \frac{c}{n}, \quad (178)$$

with the **index of refraction**

$$n = \sqrt{\frac{\varepsilon\mu}{\varepsilon_0\mu_0}} \approx \sqrt{\varepsilon_r}, \quad \varepsilon_r = 1 + \chi_e. \quad (179)$$

2. **Conductivity** σ ($= 1/\varrho$, where ϱ is the resistivity), which in general is a tensor σ_{ij} .

ε , μ , and σ all depend on the material. In addition, they depend on the **wavelength** (or energy) — and with them, the refractive index depends on energy. This is a well-known phenomenon in optics called **dispersion**.

Example: typical glass

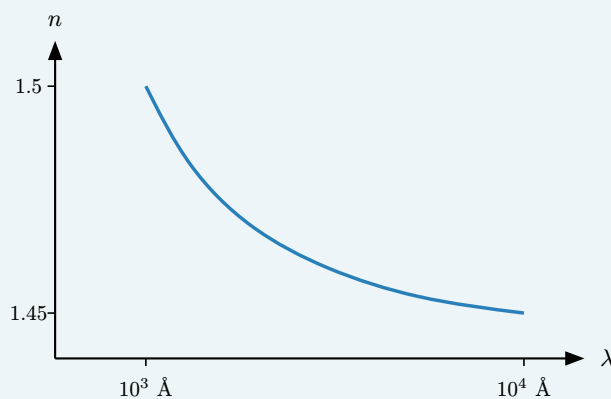


Figure 18: Index of refraction of typical glass as a function of wavelength.

This is how a prism works, and how a rainbow is formed by rain droplets.

We will also see **anomalous dispersion**, where the material experiences a resonance.

5.3.2 Phase and Group Velocity

In a dispersive medium, the wave velocity depends on the EM frequency, $v(\omega)$ (as it turns out, in quantum electrodynamics even the vacuum is dispersive!).

Each sinusoidal component travels at the **wave** (or **phase**) **velocity**

$$v = \frac{\omega}{k}, \quad (180)$$

whereas the envelope of a wave packet moves at the **group velocity**

$$v_g = \frac{d\omega}{dk}. \quad (181)$$

5.3.3 A Model for the Frequency-Dependent Permittivity

We now want to develop a model for the frequency dependence of ϵ in **dielectrics**, i.e. non-conducting media in which the electrons are bound to atoms or molecules.

We describe the binding force with a **spring force**:

$$\mathbf{F}_{\text{binding}} = -k_{\text{spring}}\mathbf{x} = -m\omega_0^2\mathbf{x}, \quad (182)$$

where x is the displacement of the electron from its equilibrium position relative to the nucleus, Figure 19, and

$$\omega_0 = \sqrt{\frac{k_{\text{spring}}}{m}}. \quad (183)$$

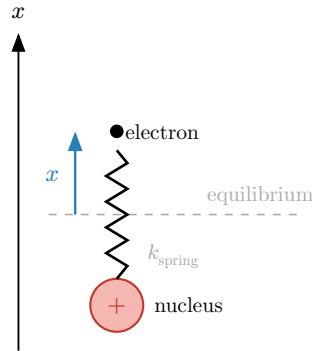


Figure 19: An electron bound to a nucleus by a spring force. The displacement x is measured from the equilibrium position.

Why a spring force is not a crazy model

If $U(x)$ was the potential of the (much more complicated) true binding force, we could expand it around the equilibrium point $x = 0$ with a Taylor series:

$$\begin{aligned} U(x) = & \underbrace{U(0)}_{\text{can be chosen arbitrarily}} + x \underbrace{U'(0)}_{=-F=0 \text{ (equilibrium)}} \\ & + \frac{1}{2}x^2 \underbrace{U''(0)}_{=k_{\text{spring}} > 0 \text{ (stable equilibrium)}} + \dots \end{aligned} \quad (184)$$

Choosing $U(0) = 0$ and using $U'(0) = 0$, this becomes

$$U(x) = \frac{1}{2}U''(0)x^2 + O(x^3), \quad (185)$$

which is exactly the potential of a harmonic oscillator, with $k_{\text{spring}} \equiv U''(0) > 0$ (we will use the same line of argument in quantum mechanics).

We probably also have a **damping force** that acts against the velocity of the electron. The simplest model is

$$\mathbf{F}_{\text{damping}} = -m\gamma \frac{d\mathbf{x}}{dt}. \quad (186)$$

And lastly, we have a **driving force** given by the electromagnetic wave, polarized in the x -direction:

$$\mathbf{F}_{\text{driving}} = q\mathbf{E} = qE_0 \cos(\omega t). \quad (187)$$

The electron is then driven by the wave, Figure 20, which combines all three forces on the bound electron.

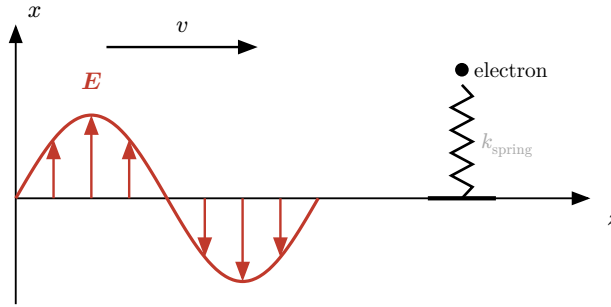


Figure 20: An electromagnetic wave (electric field $\mathbf{E}$ along x , propagating along z with velocity v) drives an electron bound to the axis by a spring.

Using Newton's second law,

$$m \frac{d^2x}{dt^2} = F_{\text{binding}} + F_{\text{damping}} + F_{\text{driving}}, \quad (188)$$

we find

$$m \frac{d^2x}{dt^2} + m\gamma \frac{dx}{dt} + m\omega_0^2 x = qE_0 \cos(\omega t), \quad (189)$$

which is a **damped harmonic oscillator**, driven at frequency ω .

This is easier to solve in complex notation, i.e. Eq. (189) is the real part of

$$\frac{d^2\tilde{x}}{dt^2} + \gamma \frac{d\tilde{x}}{dt} + \omega_0^2 \tilde{x} = \frac{q}{m} E_0 e^{-i\omega t}, \quad (190)$$

with $\tilde{x} = x + iy$.

In steady state, the system will oscillate at the driving frequency,

$$\tilde{x}(t) = \tilde{x}_0 e^{-i\omega t}. \quad (191)$$

Plugging this into Eq. (190) and solving for $\tilde{x}_0$, we find

$$\tilde{x}_0 = \frac{q/m}{\omega_0^2 - \omega^2 - i\gamma\omega} E_0. \quad (192)$$

In the end, we are interested in the polarization $\mathbf{P}$, which is the dipole moment per unit volume. Therefore, we calculate the dipole moment of one electron, which will be the real part of the following expression:

$$\tilde{p}(t) = q\tilde{x}(t) = \frac{q^2/m}{\omega_0^2 - \omega^2 - i\gamma\omega} E_0 e^{-i\omega t}. \quad (193)$$

The complex number in the denominator of Eq. (193) means that the dipole is **out of phase** with $\mathbf{E}$: it lags behind by an angle

$$\phi = \tan^{-1} \frac{\gamma\omega}{\omega_0^2 - \omega^2}. \quad (194)$$

Where does this phase come from?

$\tilde{p}$ has the form $\tilde{p} = \frac{A}{x-iy} e^{-i\omega t}$, with (comparing to Eq. (193)) $x = \omega_0^2 - \omega^2$, $y = \gamma\omega$, $A = q^2 E_0/m$.

Let's bring this to the form $\tilde{p} = \tilde{p}_0 e^{i(\phi - \omega t)}$ by multiplying numerator and denominator by the complex conjugate of the denominator:

$$\tilde{p} = \frac{A(x+iy)}{(x-iy)(x+iy)} e^{-i\omega t} = \frac{A}{x^2 + y^2} (x+iy) e^{-i\omega t}. \quad (195)$$

With $r^2 \equiv x^2 + y^2$ and $x+iy \equiv r e^{i\phi}$ (as sketched in the complex plane, with $\tan \phi = y/x$), this becomes

$$\tilde{p} = \frac{A}{r} e^{i(\phi - \omega t)}. \quad (196)$$

So the dipole oscillates at the same frequency ω as the driving field, but with an extra phase ϕ . Using $y = \gamma\omega$ and $x = \omega_0^2 - \omega^2$ from Eq. (193), we recover

$$\tan \phi = \frac{y}{x} = \frac{\gamma\omega}{\omega_0^2 - \omega^2}, \quad (197)$$

i.e. Eq. (194). ■

5.3.4 From One Electron to the Polarization $\mathbf{P}$

Let's consider a medium made up of N molecules per unit volume, where each molecule has f_j electrons. Each electron may have a different (resonant) frequency ω_j and damping γ_j .

The polarization $\mathbf{P}$ is then the real part of

$$\tilde{\mathbf{P}}(t) = \frac{Nq^2}{m} \left(\sum_j \frac{f_j}{\omega_j^2 - \omega^2 - i\gamma_j\omega} \right) \tilde{\mathbf{E}}. \quad (198)$$

In-class exercise

Previously, we considered the *real* case, where $\mathbf{E}$ and $\mathbf{P}$ were proportional to each other in linear media, $\mathbf{P} = \varepsilon_0 \chi_e \mathbf{E}$. Here, this is not the case — why?

The reason: $\tilde{\mathbf{P}}(t)$ and $\tilde{\mathbf{E}}(t)$ have a **phase difference** (Eq. (194)), so their *real parts* $\mathbf{P}(t)$ and $\mathbf{E}(t)$ are not proportional at every instant in time. But the *complex* $\tilde{\mathbf{P}}$ and $\tilde{\mathbf{E}}$ are proportional:

$$\tilde{\mathbf{P}} = \varepsilon_0 \tilde{\chi}_e \tilde{\mathbf{E}}, \quad (199)$$

with the **complex susceptibility** $\tilde{\chi}_e$.

All of our previous considerations for linear media carry over, and we can define the **complex permittivity** $\tilde{\varepsilon} = \varepsilon_0(1 + \tilde{\chi}_e)$ and the **complex dielectric constant**

$$\tilde{\varepsilon}_r = \frac{\tilde{\varepsilon}}{\varepsilon_0} = 1 + \frac{Nq^2}{m\varepsilon_0} \sum_j \frac{f_j}{\omega_j^2 - \omega^2 - i\gamma_j\omega}. \quad (200)$$

For $\gamma_j = 0$, we see that there's an infinity when $\omega \rightarrow \omega_j$ in $\tilde{\varepsilon}_r$. We call this a **resonance** (also in the case when $\gamma_j \neq 0$, where the divergence is tamed but there is still a pronounced peak).

The (complex) wave equation in a dispersive medium is (using $\mu \approx \mu_0$)

$$\nabla^2 \tilde{\mathbf{E}} = \tilde{\varepsilon} \mu_0 \frac{\partial^2 \tilde{\mathbf{E}}}{\partial t^2}, \quad (201)$$

which is solved by plane waves

$$\tilde{\mathbf{E}}(z, t) = \tilde{\mathbf{E}}_0 e^{i(\tilde{k}z - \omega t)}. \quad (202)$$

Just as with conductors, the wavenumber is complex, $\tilde{k} = k + i\kappa$, so

$$\tilde{\mathbf{E}}(z, t) = \tilde{\mathbf{E}}_0 e^{-\kappa z} e^{i(kz - \omega t)}, \quad (203)$$

but now the attenuation is not related to the conductivity, but to the *damping force* acting on the electrons.

The intensity is $I = \tilde{\mathbf{E}} \cdot \tilde{\mathbf{E}}^* \propto e^{-2\kappa z}$, and we call

$$\alpha \equiv 2\kappa \quad (204)$$

the **absorption coefficient**.

The last thing we want to do is find α and the index of refraction,

$$n = \frac{ck}{\omega} = \sqrt{\frac{\varepsilon\mu}{\varepsilon_0\mu_0}} \approx \sqrt{\varepsilon_r} = \sqrt{1 + \chi_e}. \quad (205)$$

So we need the real and imaginary parts of

$$\tilde{k} = \frac{\omega}{c} \sqrt{\tilde{\varepsilon}_r} = \frac{\omega}{c} \sqrt{1 + \tilde{\chi}_e}. \quad (206)$$

Since $\tilde{\chi}_e$ is typically small, we Taylor-expand $\sqrt{1 + \tilde{\chi}_e}$ around $\tilde{\chi}_e = 0$, keeping only the linear term:

$$\tilde{k} = \frac{\omega}{c} \sqrt{1 + \tilde{\chi}_e} \approx \frac{\omega}{c} \left(1 + \frac{1}{2} \tilde{\chi}_e \right), \quad (207)$$

$\uparrow$
 $\sqrt{1 + \tilde{\chi}_e} \approx 1 + \frac{1}{2} \tilde{\chi}_e$

and using Eq. (200) for $\tilde{\chi}_e$:

$$\tilde{k} \approx \frac{\omega}{c} \left[1 + \frac{Nq^2}{2m\varepsilon_0} \sum_j \frac{f_j}{\omega_j^2 - \omega^2 - i\gamma_j\omega} \right]. \quad (208)$$

The **index of refraction** is

$$n = \frac{c}{\omega} \operatorname{Re}(\tilde{k}) \approx \operatorname{Re} \left(1 + \frac{1}{2} \tilde{\chi}_e \right). \quad (209)$$

To extract the real part, we rationalize the denominator of each term in Eq. (208) by multiplying by the complex conjugate:

$$\frac{1}{\omega_j^2 - \omega^2 - i\gamma_j\omega} = \frac{\omega_j^2 - \omega^2 + i\gamma_j\omega}{(\omega_j^2 - \omega^2)^2 + \gamma_j^2\omega^2}, \quad (210)$$

so that

$$n \approx 1 + \frac{Nq^2}{2m\varepsilon_0} \sum_j \frac{f_j(\omega_j^2 - \omega^2)}{(\omega_j^2 - \omega^2)^2 + \gamma_j^2\omega^2}. \quad (211)$$

And the **absorption coefficient** is

$$\alpha = 2\kappa = 2 \operatorname{Im}(\tilde{k}) \approx 2 \frac{\omega}{c} \operatorname{Im} \left(1 + \frac{1}{2} \tilde{\chi}_e \right) = \frac{\omega}{c} \operatorname{Im}(\tilde{\chi}_e). \quad (212)$$

Using the imaginary part of the rationalized fraction above (i.e. $\gamma_j\omega$ in the numerator, instead of $\omega_j^2 - \omega^2$):

$$\alpha \approx \frac{Nq^2\omega^2}{m\varepsilon_0 c} \sum_j \frac{f_j\gamma_j}{(\omega_j^2 - \omega^2)^2 + \gamma_j^2\omega^2}. \quad (213)$$

In-class exercise

Check this yourself: verify the rationalization step above, and that you recover Eq. (211) and Eq. (213).

Let's sketch α and n in the vicinity of one of the resonances ω_j , see Figure 21.

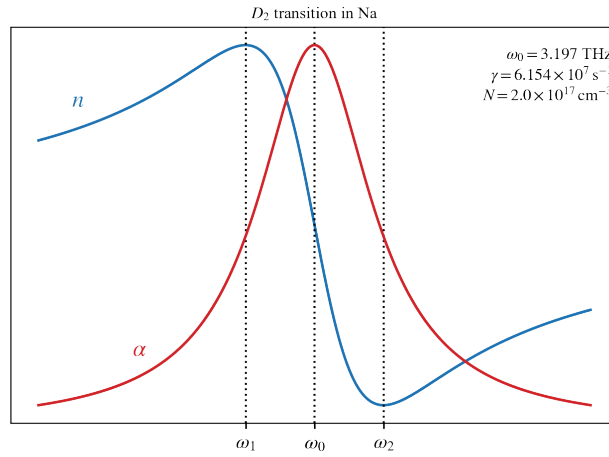


Figure 21: Index of refraction n and absorption coefficient α near the D_2 resonance of sodium.

Away from the resonance ($\omega < \omega_1$ or $\omega > \omega_2$), n increases slowly with ω — this is the behavior we saw in e.g. glass (Figure 18), called **normal dispersion**.

In the region $\omega_1 < \omega_j < \omega_2$ (surrounding ω_j), we instead see a rapid *drop* of n with increasing ω — called **anomalous dispersion** — together with a pronounced peak in the absorption α_j : the electrons are being driven at (close to) their “favorite” frequency ω_j , leading to oscillations with large amplitude, and hence a maximum amount of energy dissipated through damping. The material can become **opaque** in this region.